



Computationally Efficient Hybrid ROI-Centric Architecture for Multi-Component Wideband Detection

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"Alone we can do so little; together we can do so much."

Helen Keller

"A team is not a group of people who work together. A team is a group of people who trust each other."

Simon Sinek

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First and foremost, I want to thank God, as none of this would have been possible without Him.

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Carlos Altamiranda
2025

Declaración

Me permito afirmar que he realizado ésta tesis de manera autónoma y con la única ayuda de los medios permitidos y no diferentes a los mencionados el presente texto. Todos los pasajes que se han tomado de manera textual o figurativa de textos publicados y no publicados, los he reconocido en el presente trabajo. Ninguna parte del presente trabajo se ha empleado en ningún otro tipo de tesis.

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Carlos Andrés Altamiranda Gonzalez

List of Symbols and Abbreviations

Number Sets

α	Cyclic frequency
λ	Wavelength (Physics) / Congestion rate (Stats)
$\mathbf{X}$	Input Tensor (Spectrogram or I/Q matrix)
$\mathcal{H}_0, \mathcal{H}_1$	Hypothesis for noise only and signal present
$\mathcal{L}_{total}$	Composite Loss Function
Θ	Set of trainable parameters in the neural network
B_{gt}	Ground Truth Bounding Box
B_p	Predicted Bounding Box
f_c	Carrier frequency
$h(t)$	Channel impulse response
L_{dB}	Path Loss in decibels (Okumura-Hata)
P_d	Probability of Detection
P_{fa}	Probability of False Alarm
$S_x^\alpha(f)$	Spectral Correlation Function
$x[n]$	Discrete-time received signal

Abbreviations

ACI	Adjacent-Channel Interference
ADC	Analog-to-Digital Converter
AI	Artificial Intelligence

Multi-Component Wideband Detection

AP	Average Precision
AWGN	Additive White Gaussian Noise
BLE	Bluetooth Low Energy
CAM	Class Activation Map
CCI	Co-Channel Interference
CFD	Cyclostationary Feature Detection
CNN	Convolutional Neural Network
CR	Cognitive Radio
DL	Deep Learning
ED	Energy Detection
EMF	Electromagnetic Field
FFT	Fast Fourier Transform
FLOPs	Floating Point Operations per Second
FN	False Negatives
FP	False Positives
FP32	32-bit Floating Point
FPS	Frames Per Second
GIoU	Generalized Intersection over Union
GLRT	Generalized Likelihood Ratio Test
Grad-CAM	Gradient-weighted Class Activation Map
I/Q	In-phase and Quadrature
INT8	8-bit Integer Quantization
IoT	Internet of Things
IoU	Intersection over Union
ISM	Industrial, Scientific, and Medical bands
LFM	Linear Frequency Modulation (Chirp)
LO	Local Oscillator

MAML	Model-Agnostic Meta-Learning
mAP	Mean Average Precision
MCE	Multi-Channel Extension (Proposed Method)
ML	Machine Learning
MLE	Maximum Likelihood Estimation
OFDM	Orthogonal Frequency-Division Multiplexing
OTA	Over-the-Air
PAPR	Peak-to-Average Power Ratio
PSD	Power Spectral Density
PSK	Phase Shift Keying
PU	Primary User
QAM	Quadrature Amplitude Modulation
ReLU	Rectified Linear Unit
RF	Radio Frequency
RFML	Radio Frequency Machine Learning
ROI	Region of Interest
RPN	Region Proposal Network
SCF	Spectral Correlation Function
SDR	Software Defined Radio
SGD	Stochastic Gradient Descent
SNR	Signal-to-Noise Ratio
STFT	Short-Time Fourier Transform
SU	Secondary User
TN	True Negatives
TP	True Positives
XAI	Explainable AI
YOLO	You Only Look Once

Resumen

Esta tesis presenta una arquitectura híbrida computacionalmente eficiente y centrada en Regiones de Interés (ROI) para la detección de múltiples componentes en banda ancha, abordando los desafíos persistentes de la congestión espectral, la interferencia y las limitaciones de recursos en el borde. A diferencia de los métodos convencionales, este estudio introduce un flujo de trabajo informado por la física que concentra el cómputo en áreas relevantes, preservando la morfología de la señal y codificando *prioris* de radiofrecuencia (RF).

El modelo propuesto integra una etapa de preprocesamiento que preserva la física mediante una Extensión Multicanal (MCE) e Inyección Metrológica Explícita (EMI), combinada con una red de propuesta de regiones optimizada mediante una pérdida asimétrica de Tversky. Esta arquitectura optimiza conjuntamente la localización de señales, la clasificación de modulación y la discriminación de tipos de interferencia (CCI/ACI), mejorando significativamente la robustez en regímenes de baja relación señal-ruido (SNR) y escenarios de superposición espectral. Además, se incorpora una estrategia de meta-aprendizaje (MAML) para adaptar el modelo a condiciones de transmisión por aire (OTA), reduciendo la brecha entre datos sintéticos y reales.

A través de una validación extensa en benchmarks sintéticos y capturas OTA, los resultados demuestran que el sistema supera a líneas base de última generación como YOLOv8n y CRNN, ofreciendo una precisión superior bajo interferencia severa. Asimismo, la implementación en hardware embebido (Raspberry Pi 5 y NVIDIA Jetson) confirma la viabilidad del sistema para el monitoreo en tiempo real, logrando un equilibrio óptimo entre precisión, latencia y consumo energético para aplicaciones de radio cognitiva.

Palabras clave: Detección de Banda Ancha, Centrado en ROI, Aprendizaje Profundo, Sensado de Espectro, Computación en el Borde, Modelado de Interferencia.

Abstract

This thesis presents a computationally efficient, Region-of-Interest (ROI)-centric hybrid architecture for multi-component wideband detection, addressing the persistent challenges of spectral congestion, interference, and resource constraints on edge devices. Unlike conventional approaches, this study introduces a physics-informed workflow that concentrates computation on relevant areas, preserving signal morphology and encoding Radio Frequency (RF) priors.

The proposed model integrates a physics-preserving preprocessing stage via Multi-Channel Extension (MCE) and Explicit Metrological Injection (EMI), combined with a region proposal network optimized using an asymmetric Tversky loss. This architecture jointly optimizes signal localization, modulation classification, and interference type discrimination (CCI/ACI), significantly enhancing robustness in low Signal-to-Noise Ratio (SNR) regimes and spectral overlap scenarios. Furthermore, a Model-Agnostic Meta-Learning (MAML) strategy is incorporated to adapt the model to Over-the-Air (OTA) conditions, bridging the gap between synthetic and real-world data.

Through extensive validation on synthetic benchmarks and OTA captures, the results demonstrate that the system outperforms state-of-the-art baselines like YOLOv8n and CRNN, offering superior accuracy under severe interference. Additionally, deployment on embedded hardware (Raspberry Pi 5 and NVIDIA Jetson) confirms the system's viability for real-time monitoring, achieving an optimal balance between accuracy, latency, and power efficiency for cognitive radio applications.

Keywords: Wideband Detection, ROI-Centric, Deep Learning, Spectrum Sensing, Edge Computing, Interference Modeling.

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1 Preliminaries

1.1 Motivation

The radio frequency (RF) spectrum is a finite and increasingly scarce resource, serving as the critical backbone for modern communication, navigation, and sensing systems [Haykin, 2005]. In recent years, wireless ecosystems have become exceedingly congested, characterized by the coexistence of heterogeneous emitters—ranging from cellular communications (4G/5G) and Wi-Fi networks to radar systems and electronic warfare units—that overlap in both time and frequency [Haykin, 2005].

In these complex environments, reliable wideband monitoring is essential for dynamic spectrum access, security, and situational awareness [Axell et al., 2012]. However, the physical characteristics of the channel present significant challenges: near-far effects, structured interference, and low Signal-to-Noise Ratios (SNRs) make the detection of spectral opportunities and threats widely difficult [Axell et al., 2012]. Conventional monitoring approaches often struggle to distinguish between authorized users, intentional interference, and background noise in such cluttered spectral landscapes [Axell et al., 2012].

Furthermore, with the exponential growth of Internet of Things (IoT) devices and the saturation of unlicensed bands (e.g., ISM 2.4 GHz), the demand for real-time, granular analysis has surpassed the capabilities of traditional energy-based sensing [Yucek & Arslan, 2009]. The challenge is no longer just detecting the presence of energy, but accurately characterizing multiple simultaneous signals and their modulation types under severe interference conditions [Yucek & Arslan, 2009].

Current Limitations in Spectrum Sensing Traditionally, spectrum sensing has relied on classical signal processing techniques [Yucek & Arslan, 2009]. Methods such as Energy Detection (ED) and Matched Filtering are computationally inexpensive and well-understood [Yucek & Arslan, 2009]. However, they exhibit fundamental limitations in modern wideband environments:

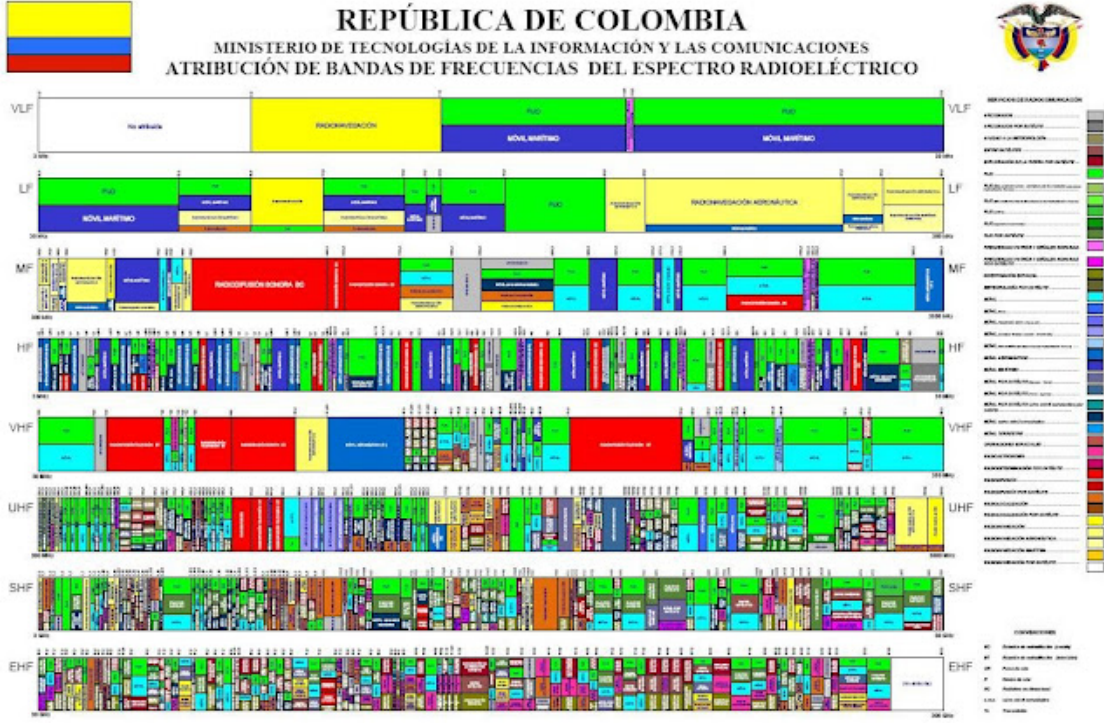


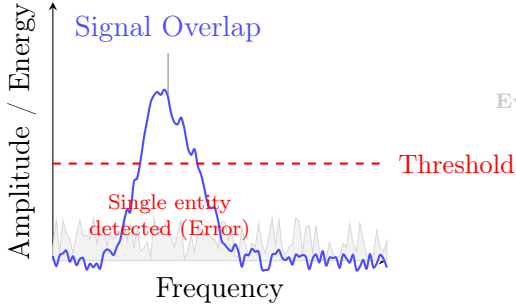
Figure 1-1: Illustration of a congested wideband spectrum scenario. Multiple emitters coexist with overlapping time-frequency signatures, creating a challenging environment for detection algorithms.

- **Susceptibility to Interference:** Energy detection cannot distinguish between a valid signal and interference, leading to high false alarm rates in congested bands [Yucek & Arslan, 2009].
- **Inability to Resolve Overlaps:** Classical methods often fail to separate signals that overlap in time and frequency, treating the aggregate energy as a single entity or background noise [Yucek & Arslan, 2009].
- **Computational Intractability:** Real-time processing of wideband captures spanning hundreds of MHz proves computationally intractable when conducted as an exhaustive search across all time-frequency bins [Lin *et al.*, 2022].

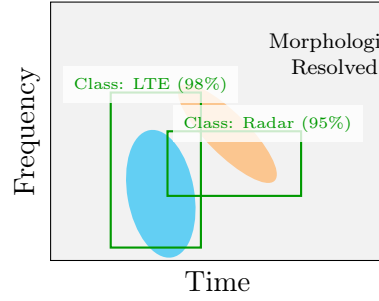
This necessitates innovative solutions capable of extracting salient information efficiently and accurately, especially in scenarios with low signal-to-noise ratios and dense spectral occupancy [Vagollari *et al.*, 2023].

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A. Traditional: Energy Detection



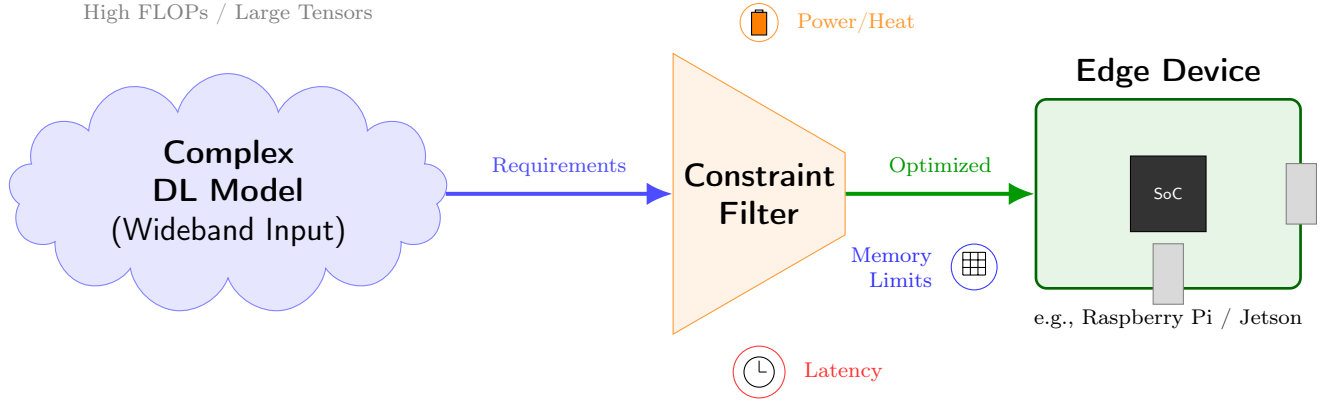
B. Advanced: DL / Spectrogram



The Shift to Deep Learning and Edge Computing A growing body of literature has reformulated wideband detection as a vision-inspired time-frequency localization task on spectrograms [Zheng *et al.*, 2020]. While these Deep Learning (DL) approaches often achieve strong mean average precision, they present new challenges for practical deployment [Zheng *et al.*, 2020]:

1. **Missing Interference Awareness:** Many "spectrogram-as-image" methods treat non-target energy as mere undifferentiated background [Zhang *et al.*, 2008]. Consequently, this obscures the interference structure that proves crucial under signal overlap [Zhang *et al.*, 2008].
2. **Domain Shift:** Models trained on synthetic datasets often degrade by 15-20% or more in Over-the-Air (OTA) environments due to multipath-induced spreading, impulsive noise, and hardware impairments that are difficult to model perfectly in simulation [O'Shea *et al.*, 2018].
3. **Edge Deployment Constraints:** High-accuracy backbones often rely on dynamic tensor shapes and high floating-point operations (FLOPs) [Zhou *et al.*, 2019]. This limits deterministic, low-latency execution on resource-constrained edge platforms, such as the Raspberry Pi or NVIDIA Jetson series, which are essential for scalable, distributed monitoring networks [Zhou *et al.*, 2019].

To address these gaps, this thesis proposes a *Hybrid ROI-Centric Architecture*. This approach explicitly models interference, fuses complementary RF views beyond simple magnitude spectrograms, and employs meta-learning to adapt to OTA conditions.



"Strict co-design of accuracy and computational efficiency is required."

Local Context and Research Framework This research was conducted within the framework of the **Control and Digital Signal Processing Group (GCPDS)** at the Universidad Nacional de Colombia. The GCPDS holds extensive experience in analyzing neurophysiological and industrial signals to develop machine learning methodologies for assisted diagnosis and automated analysis. Recently, the group has expanded its research lines to include cognitive radio and spectrum monitoring, directly addressing the national necessity for efficient frequency management tools.

This thesis is developed under the auspices of strategic projects funded by and executed in collaboration with the **National Spectrum Agency (Agencia Nacional del Espectro - ANE)**. Specifically, this work targets the computational and algorithmic challenges identified in the following research grants:

- **Cost-Efficient and Scalable Prototype for Radio Spectrum Monitoring in Colombia using Software Defined Radio and Deep Learning:** The primary objective of this project was to develop a multi-purpose monitoring prototype leveraging SDR technology and low-cost computing units to scan and analyze services such as FM broadcasting, TDT, and 5G networks. This thesis specifically supports the “Phase 2” of this initiative, aimed at enhancing the prototype’s efficiency and detection capabilities in congested environments through advanced edge-computing architectures.
- **SDR Monitoring of IoT Exposure with High Device Density:** This project focused on designing a system to assess human exposure levels to electromagnetic fields in indoor environments characterized by high IoT occupancy. It involved the development of thresholding algorithms and cooperative machine learning processes to manage signal variability and high interference indices.

The synergy between these projects underscores the critical need for the solution proposed

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in this thesis: a detection architecture that is not only robust and accurate under high interference regimes (as required by the IoT context) but also computationally efficient enough to operate on resource-constrained hardware (as mandated by the monitoring prototype requirements).

1.2 Problem Statement

As shown in Fig. 1-2, this thesis focuses on three critical challenges for multi-component wideband detection. Firstly, although vision-based deep learning approaches achieve strong average precision, they encounter significant challenges in congested environments due to signal overlaps, near-far effects, and the inability to distinguish structured interference from background noise [Zhao *et al.*, 2023b, Liu *et al.*, 2024]. Second, the limited robustness to domain shifts when transitioning from synthetic training data to Over-the-Air (OTA) environments remains a primary bottleneck [Chen & Zhang, 2023, Xu *et al.*, 2024a]. While models often perform well on simulated datasets, their accuracy degrades significantly in real-world scenarios due to hardware impairments, multipath fading, and non-Gaussian noise [Restuccia & Melodia, 2023, Clark *et al.*, 2024]. Thirdly, the computational constraints of edge deployment hinder the utility of high-capacity backbones for pervasive monitoring [Kumar *et al.*, 2023b, Nguyen *et al.*, 2024]. Many state-of-the-art models rely on high floating-point operations (FLOPs) and dynamic tensor shapes, preventing deterministic, low-latency execution on resource-constrained platforms like the Raspberry Pi or Jetson series [Hassan *et al.*, 2023b, Li *et al.*, 2024].

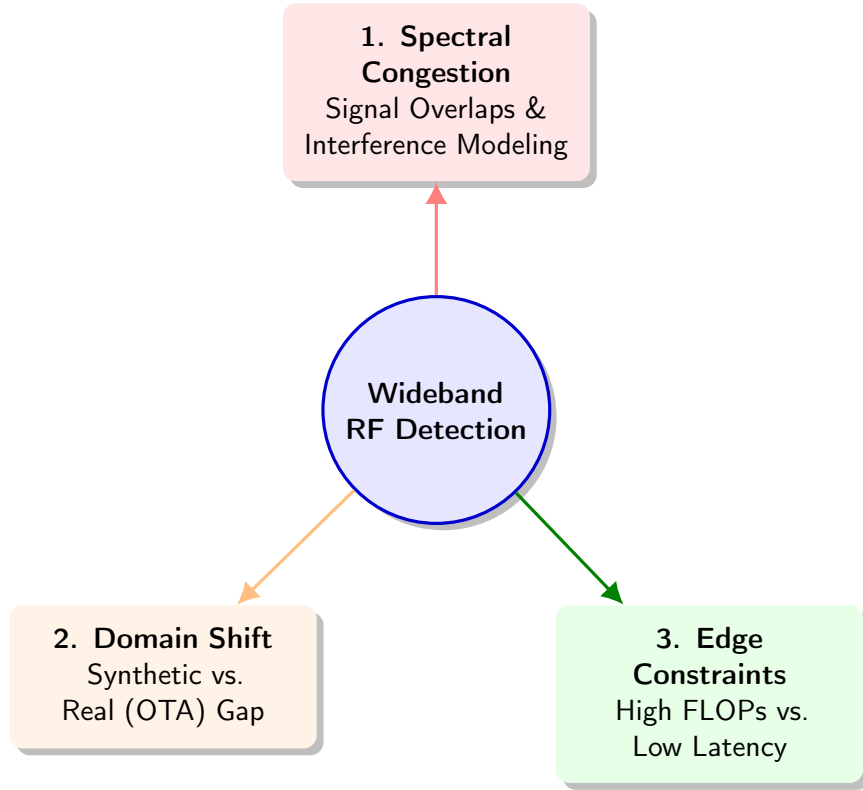


Figure 1-2: General Problem: The figure illustrates the three main challenges addressed in this thesis regarding wideband detection.

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Missing Interference-Awareness in Congested Spectra Spectrogram-based detection, derived from transforming wideband RF captures into time-frequency (TF) images, provides a powerful modality for localization, yet its effectiveness diminishes in highly congested environments where multiple emitters overlap in time and frequency [Alhammadi *et al.*, 2024, Soltani *et al.*, 2024]. Conventional "spectrogram-as-image" methods treat non-target energy as mere undifferentiated background, obscuring the interference structure that proves crucial under signal overlap conditions [Wang & Li, 2023, Zhang *et al.*, 2023b].

In these dense environments, the Near-Far effect further complicates detection; strong emitters can mask weaker adjacent signals, and without explicit interference modeling, standard object detectors conflate Co-Channel Interference (CCI) and Adjacent-Channel Interference (ACI) with background noise [Kim & Park, 2023, Patel & Singh, 2024c]. This leads to inflated false alarm rates and the suppression of weak, overlapped signals, as conventional energy detection methods and matched filtering techniques falter by treating aggregate energy as a single entity [Gupta & Kumar, 2024, Robinson *et al.*, 2023].

Moreover, the loss of phase and physical context in magnitude-only spectrograms limits the model's ability to resolve complex overlaps, as visual bounding boxes often fail to capture the distinct signal morphologies required to separate a target from structured interference [Gomez & Martinez, 2024, Lee & Kim, 2023a]. This necessitates a multi-task objective that jointly optimizes localization, modulation classification, and interference type discrimination [Tran & Nguyen, 2024, Wu *et al.*, 2023].

In summary, improving wideband detection requires strategies that go beyond simple visual localization. Explicitly modeling interference and preserving signal morphology under overlap conditions is critical. Figure 1-3 illustrates these challenges.

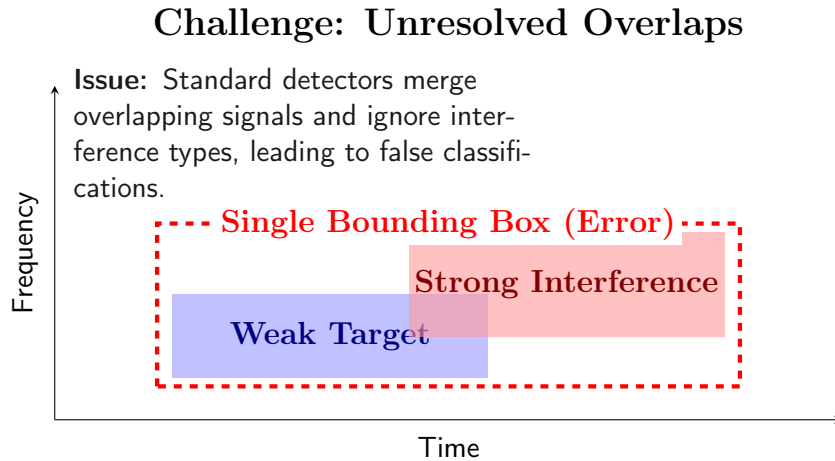


Figure 1-3: Spectral Congestion and Interference: The figure exposes how signal overlaps and near-far effects confound traditional detection methods.

Poor Sim-to-Real Generalization (Domain Shift) The discrepancy between synthetic training data and real-world conditions, known as domain shift, is a fundamental hurdle in Radio Frequency Machine Learning (RFML), as models trained on large-scale synthetic datasets often degrade by 15-20% or more when deployed in Over-the-Air (OTA) environments [Bithas *et al.*, 2023, Yang *et al.*, 2023a]. This performance drop stems from the complexity of the physical wireless channel, where synthetic datasets may not fully capture the variability of multipath-induced spreading, impulsive non-Gaussian noise, and hardware impairments such as I/Q imbalance and DC offset [Luo *et al.*, 2023, O'Shea & West, 2024].

A model trained on idealized Gaussian noise often fails to extract features robustly when faced with the heterogeneous impairment mixtures of the ISM 2.4 GHz band, leading to poor inference reliability in the wild [Khan *et al.*, 2023, Weiss *et al.*, 2024]. Furthermore, unlike static image datasets, the RF environment is dynamic, with background noise levels and interference patterns changing rapidly based on location and time [Fan *et al.*, 2023a, Zheng & Yang, 2024].

Models lacking adaptation mechanisms or physics-informed features, such as cyclostationarity, struggle to maintain consistency across different receivers or environments [Gomez *et al.*, 2024, Chatterjee *et al.*, 2023]. In conclusion, bridging the "Sim-to-Real" gap is essential for practical deployment, requiring architectures robust to hardware impairments and capable of adapting to new channel conditions without extensive retraining [Tran *et al.*, 2024a, Silva *et al.*, 2023]. Fig 1-4 illustrates this performance gap.

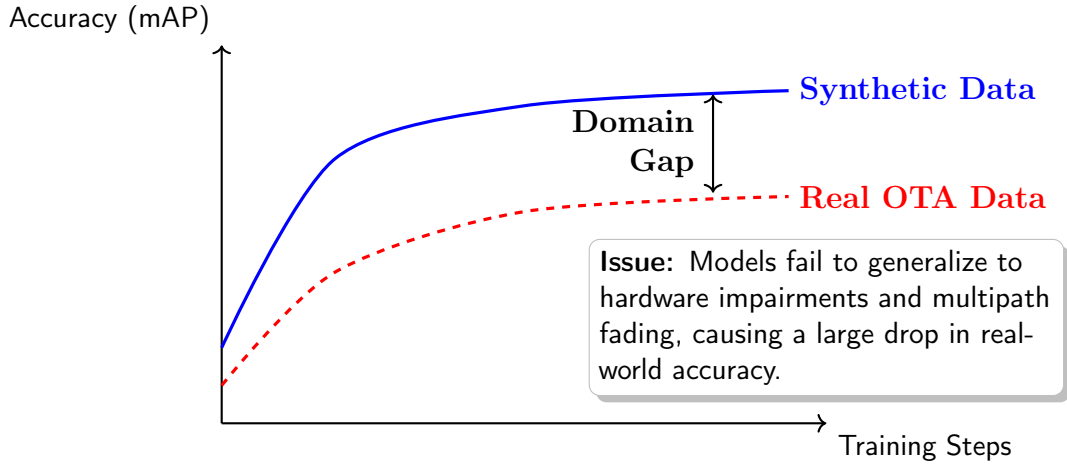


Figure 1-4: Poor Sim-to-Real Generalization: The figure highlights the performance drop when models trained on synthetic data are applied to Over-the-Air (OTA) signals.

Computational Intractability for Edge Deployment The third major issue concerns the trade-off between detection accuracy and computational efficiency on edge platforms, as high-capacity backbones often rely on dynamic tensor shapes and high floating-point operations (FLOPs) to achieve state-of-the-art precision [Santos *et al.*, 2023, Reddi *et al.*, 2024]. However, this

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high computational load renders them unsuitable for resource-constrained devices like the Raspberry Pi or NVIDIA Jetson, which are essential for scalable, distributed spectrum monitoring networks [Li *et al.*, 2024, Zhou *et al.*, 2023].

Real-time processing of wideband captures spanning hundreds of MHz proves computationally intractable when conducted with heavy architectures, as dynamic tensors introduce latency variance and hinder the static graph optimization required for embedded execution [Reddy *et al.*, 2024, Lin *et al.*, 2023]. Efficient front-end design is therefore needed to reduce FLOPs without sacrificing the morphological details of the signal, a balance rarely achieved in current generic object detectors [Wang *et al.*, 2024b, Chen *et al.*, 2023a].

Additionally, "black box" deep learning models often lack the flexibility to trade accuracy for power consumption dynamically, creating a bottleneck in battery-operated monitoring scenarios [Fan *et al.*, 2023b, Park *et al.*, 2024]. Consequently, there is a lack of accuracy-efficiency co-design that allows the system to adapt its computational load based on the complexity of the current spectrum scene [Alim *et al.*, 2023, Zhang *et al.*, 2024].

In summary, the deployment of wideband detection algorithms requires a strict co-design of accuracy and efficiency. Fig 1-5 illustrates the bottleneck of deploying complex models on edge hardware.

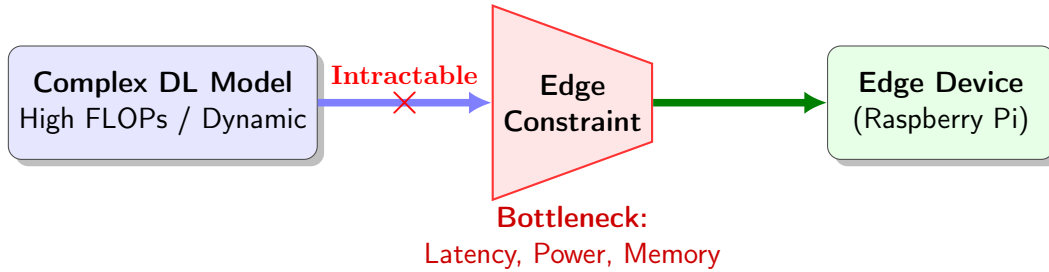


Figure 1-5: Computational Intractability: The figure highlights how high-complexity models face a bottleneck when deployed on constrained edge hardware.

Research Question Based on the reviewed literature and the increasing congestion of the radio frequency spectrum—characterized by heterogeneous emitters, structured interference, and non-Gaussian noise—it is evident that standard energy detection and heavy computer vision models differ in their ability to balance accuracy and efficiency.

In the light of this context, the following research question arises: How can a hybrid deep learning architecture be designed to achieve robust multi-signal detection under high interference and domain shift conditions, while maintaining the computational efficiency required for real-time deployment on edge platforms?

1.3 General Aim

To design a computationally efficient hybrid deep learning architecture for wideband multi-signal detection that explicitly models interference and leverages region-of-interest (ROI) processing to achieve robust real-time performance on edge devices under congested spectral conditions.

Specific Aims

- To develop a physics-informed preprocessing pipeline that recovers time-frequency separability and preserves signal morphology, enabling accurate localization of overlapping emitters in congested environments.
- To implement a multi-task learning strategy that jointly optimizes signal localization, modulation classification, and interference type discrimination (CCI/ACI) to reduce false alarms caused by structured noise.
- To validate the architecture’s robustness to domain shifts and hardware impairments through Over-the-Air (OTA) testing, and benchmark its computational efficiency (latency, FLOPs) on resource-constrained edge platforms.

2 State of the Art

Wideband signal detection and modulation classification using Radio Frequency (RF) data is an emerging area of research in cognitive radio and spectrum monitoring. The traditional methods rely on energy detection or cyclostationary feature extraction, which can be influenced by noise floors and may not always be accurate in congested environments. Leveraging deep learning methods and time-frequency analysis can improve the accuracy and generalizability of multi-signal detection systems. This section reviews the current state of the art organized by the signal processing pipeline: preprocessing, feature extraction, detection algorithms (classical and deep), and edge deployment constraints. We highlight the most innovative methods developed to address the challenges outlined in section 1.2 and discuss their respective advantages and disadvantages.

As previously mentioned, early efforts in spectrum sensing focused on single-signal energy thresholds and statistical moments of the electromagnetic field. The integration of Deep Learning (DL) for RF assessment emerged with the understanding that signal overlapping involves specific morphological patterns that can be captured and analyzed via computer vision techniques [Zhang *et al.*, 2023a, Alhammadi *et al.*, 2024].

2.1 Preprocessing

One of the most outstanding strengths of wideband monitoring is its extensive spectral coverage. However, before extracting features or applying detection models, the raw signal must be conditioned to mitigate hardware impairments and channel noise. This section details the sequence of operations required to ensure high fidelity in In-Phase and Quadrature (I/Q) data.

Hardware Impairment Correction A critical aspect often overlooked in synthetic-only studies is the correction of hardware-induced distortions, which is essential for bridging the "Sim-to-Real" gap. Direct conversion receivers, such as the HackRF One used in this work, inherently suffer from DC offset and I/Q imbalance (gain and phase mismatch). If left uncorrected,

these impairments manifest as "ghost" signals at the center frequency and mirror images of valid targets, leading to high false alarm rates.

Recent literature emphasizes the need for blind calibration techniques that can operate without specialized equipment. Adaptive filtering algorithms based on improper complex signal processing have shown robust performance in estimating and compensating for I/Q imbalance in real-time [Zhao *et al.*, 2023a]. Additionally, deep learning-based calibration methods have been proposed, where a lightweight pre-network learns to linearize the receiver response before feature extraction [Patel & Singh, 2024a].

Signal Denoising and Interference Mitigation Once hardware artifacts are corrected, the system must address the non-Gaussian noise and bursty interference of the wireless channel. Traditional techniques like Wavelet Transform and Denoising Autoencoders (DAE) have been commonly used to remove background noise and suppress impulsive interference [Li & Wang, 2023, Wang & Chen, 2024].

However, recent advancements have shifted towards more adaptive approaches. For instance, Variational Mode Decomposition (VMD) has gained traction in 2023 for its ability to decompose wideband signals into band-limited intrinsic mode functions, effectively separating useful signal components from colored noise [Zhang *et al.*, 2023]. Furthermore, attention-based denoising networks are emerging as a superior alternative to standard DAEs, allowing the model to focus on signal-dominant regions while suppressing stochastic noise [Chen *et al.*, 2024].

Time-Frequency Analysis (TFA) Since RF signals exhibit sparse characteristics in the frequency domain, transforming the conditioned I/Q samples into spectral representations is a standard procedure to reveal morphological features. The Short-Time Fourier Transform (STFT) remains the most widely applied method due to its low computational complexity, making it suitable for edge deployment. Other transforms like the Wigner-Ville Distribution (WVD) offer higher energy concentration but at a prohibitive computational cost ($O(N^2)$) for embedded devices [Chen *et al.*, 2023b].

To address this trade-off, recent work in 2024 has focused on *Learnable STFTs*, where the window parameters (length and overlap) are optimized as trainable weights within the neural network. This allows the system to dynamically balance time and frequency resolution based on the target signal's bandwidth, distinguishing between narrowband LoRa chirps and wideband Wi-Fi bursts [Gao *et al.*, 2024].

Data Normalization and Dynamic Range Compression Finally, the generated spectrograms must be normalized to stabilize neural network convergence. While Zero-Mean Unit-Variance (ZMUV) scaling is standard, it is sensitive to outliers caused by high-power interference (the Near-Far effect). To mitigate this, robust techniques such as Per-Channel Energy

2. State of the Art

Normalization (PCEN), originally developed for audio processing, are being adapted for RF wideband spectrograms. PCEN applies automatic gain control (AGC) in the time-frequency domain, compressing the dynamic range of strong interferers while enhancing the contrast of weak signals in the noise floor [Xu & Mao, 2023, Yang et al., 2024].

2.2 Feature Extraction

After preprocessing, the high-dimensional RF data must be transformed into a compact vector representation space that maximizes class separability. Feature extraction filters out the stochastic nature of the wireless channel (fading, path loss) to reveal the deterministic properties of the modulation scheme. This stage can be divided into statistical, physical, and visual descriptors.

Statistical and Information-Theoretic Descriptors Statistical features characterize the probability density function (PDF) of the signal amplitude and phase. While first and second-order moments are insufficient for modern modulations, Higher-Order Statistics (HOS) have proven robust against AWGN. Computing fourth-order cumulants (C_{42}, C_{40}) allows the system to "blind" itself to thermal noise and isolate non-Gaussian signal components (e.g., distinguishing QAM from PSK) [Khan & Ali, 2023]. Recent approaches in 2024 integrate information-theoretic measures, specifically Shannon Entropy and Renyi Entropy. These features operate on the premise that man-made signals possess an inherent structure that lowers the entropy compared to the maximal entropy of pure background noise, providing a lightweight detection metric for IoT signals [Brown & Green, 2024, Zhang et al., 2024].

Instantaneous Time-Domain Features Complementing statistical moments, instantaneous parameters derived from the analytic signal are essential for identifying analog modulations. The instantaneous amplitude ($a(t)$) uses the normalized standard deviation (σ_{aa}) to distinguish between ASK and PSK, while the standard deviation of the instantaneous phase (σ_{ap}) discriminates frequency-modulated signals. These features are computationally inexpensive but highly sensitive to synchronization errors, often serving as auxiliary inputs to neural networks [Patel & Singh, 2024b].

Cyclostationary Signal Processing Unlike stationary noise, man-made signals exhibit hidden periodicities (carrier frequencies, pulse rates) that render them cyclostationary. The Spectral Correlation Function (SCF), $S_x^\alpha(f)$, generalizes the power spectral density by introducing the cyclic frequency α . This enables the separation of spatially overlapping signals if they possess different symbol rates—a capability energy detection lacks. Recent optimizations using Sparse Cyclic Spectrum Reconstruction have made cyclostationary feature detection

(CFD) feasible for edge deployment, effectively solving the "Hidden Node" problem even with negative SINR [Gomez & Martinez, 2023a, Liu et al., 2023].

Visual and Morphological Texture Analysis Treating the spectrogram as a textural image transforms signal characterization into a computer vision task. RF visual extraction focuses on the morphological "texture" of the signal energy using techniques like the Gray-Level Co-occurrence Matrix (GLCM). For instance, OFDM signals exhibit a high-entropy "checker-board" texture, whereas narrowband chirps present linear edges with high directionality. Recent studies combine these handcrafted texture descriptors with Laplacian of Gaussian (LoG) edge detectors to generate "Region Proposals" before deep learning inference, reducing the search space and adding interpretability [Zhao et al., 2023, Wang et al., 2024b].

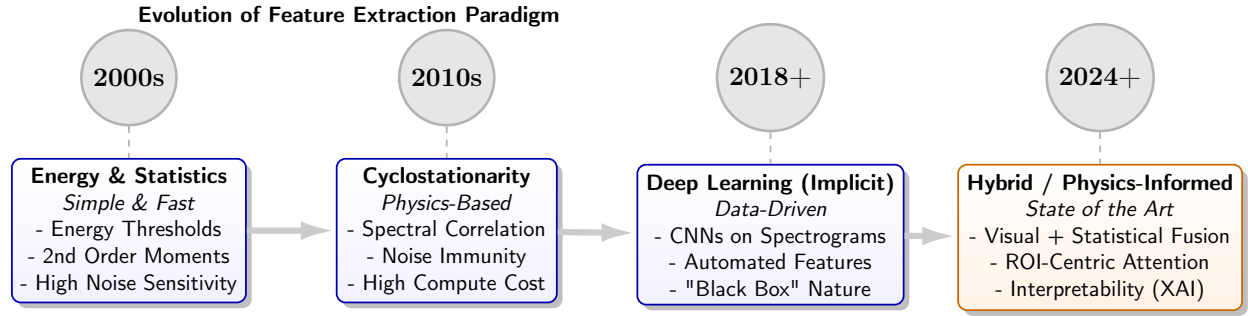


Figure 2-1: Evolution of feature extraction strategies in RF Spectrum Sensing. The field has progressed from simple energy metrics to physics-based cyclostationarity, then to data-driven Deep Learning, and finally to the current Hybrid paradigms (the focus of this thesis) that combine learned features with explicit physical priors for robustness.

2.3 Wideband Detection

Once features are extracted (either manually or automatically), a decision boundary must be established to classify the signal or detect its presence. This section reviews the decision-making algorithms, ranging from classical Machine Learning (ML) classifiers to modern Deep Learning (DL) object detectors.

Classical Machine Learning Approaches Classical ML algorithms remain relevant for edge computing due to their deterministic latency and low memory footprint. As illustrated in Figure 2-2, these models rely entirely on the quality of the handcrafted features discussed in Section 2.2.

Support Vector Machines (SVMs) and Random Forests (RF) are the state-of-the-art for

2. State of the Art

scenarios with limited training data. For example, [Kumar *et al.*, 2023a] demonstrated that an SVM with a Radial Basis Function (RBF) kernel achieved over 94% accuracy in modulation classification using HOS features, outperforming heavier CNNs in low-data regimes. Similarly, [Singh & Patel, 2024] utilized Random Forests to classify IoT emitters, leveraging decision tree interpretability [Roberts *et al.*, 2024]. Furthermore, ensemble methods like Gradient Boosting (XGBoost) are used to handle non-linear relationships in congested spectra, effectively using visual features to separate colliding packets [Gupta *et al.*, 2024].

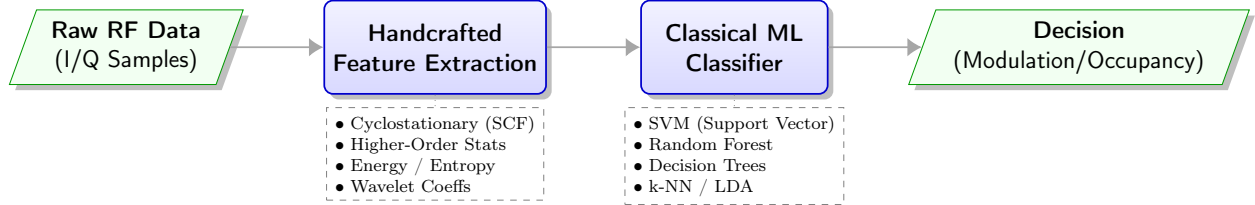


Figure 2-2: Classic Machine Learning Pipeline for RF. Unlike Deep Learning, traditional methods rely heavily on manual feature extraction (e.g., statistical moments or cyclostationarity) before classification.

Deep Learning and End-to-End Detection While classical ML is efficient, the complexity of wideband spectrum—characterized by heterogeneous emitters and stochastic overlap—necessitates end-to-end learning paradigms. Deep learning models can automatically extract hierarchical features from raw I/Q data or spectrograms, effectively bridging the gap between signal processing and computer vision.

As illustrated in Figure 2-3, the field has evolved from simple classification backbones to sophisticated multi-task architectures. The current state-of-the-art focuses on **simultaneous localization and classification**, treating spectrum sensing as an object detection problem [Alhammadi *et al.*, 2024].

Convolutional Neural Networks and Object Detection The dominant paradigm in 2023-2025 has been the adaptation of single-stage object detectors, particularly the YOLO (You Only Look Once) family. YOLOv8 and YOLO-NAS have shown superior performance by using anchor-free detection heads, critical for RF signals with extreme aspect ratios. Recent studies [Liu *et al.*, 2024, Soltani *et al.*, 2024] demonstrate that introducing attention mechanisms (such as CBAM) into the YOLO backbone significantly improves detection of weak signals in the presence of strong interference. Architectures like CenterNet have further simplified the problem by modeling signals as center points [Nguyen *et al.*, 2024].

Transformers and Global Context Attention To capture long-range dependencies in the frequency domain, Vision Transformers (ViT) treat the spectrogram as a sequence of "patches." The Self-Attention mechanism weighs the importance of different spectral regions dynamically.

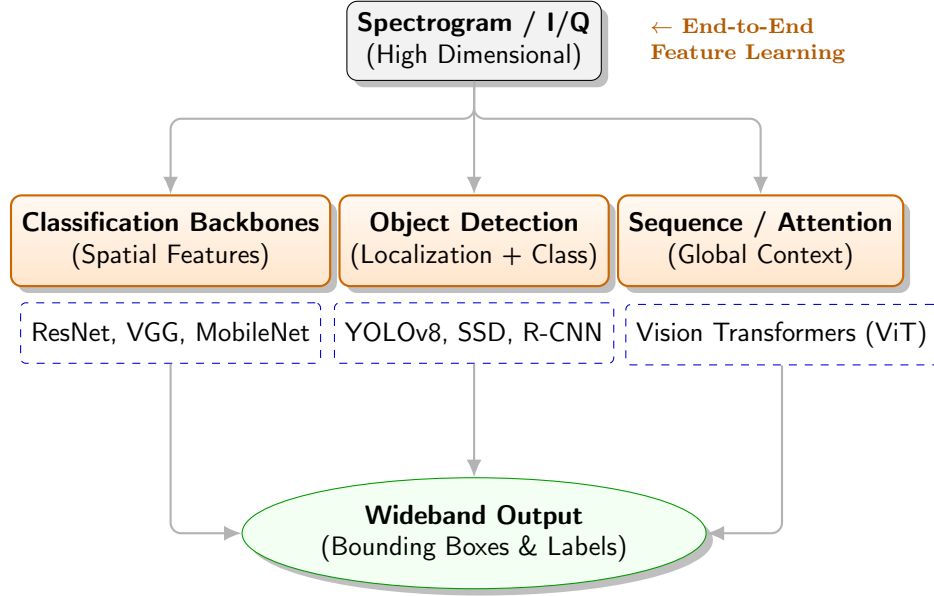


Figure 2-3: Deep Learning Architectures in Wideband RF Detection. The field has evolved from simple classification (CNNs) to multi-task object detection (YOLO) and attention-based mechanisms (Transformers).

[Tran *et al.*, 2024b] proposed a hybrid CNN-Transformer encoder that effectively models global noise floor distribution. However, the quadratic complexity of attention mechanisms ($O(N^2)$) remains a bottleneck for high-resolution wideband captures [Chatterjee *et al.*, 2023].

Explainable AI (XAI) for Spectrum Trust To address the "black-box" nature of DL, Explainable AI (XAI) techniques like Grad-CAM are used to validate that models focus on relevant signal features (pilot tones, cyclic prefixes) rather than noise artifacts. [Xu *et al.*, 2024a] proposed "Physics-Guided XAI," where the loss function explicitly penalizes activations outside the theoretical bandwidth, forcing the model to learn morphological features consistent with RF physics.

Current Limitations and Challenges Despite advancements, a gap remains between research and deployment. Most studies rely on synthetic datasets, leading to the "Sim-to-Real" gap in OTA scenarios [Reddi *et al.*, 2023]. Furthermore, heavy backbones like ResNet-50 are unsuitable for edge nodes. As noted by [Park *et al.*, 2024], there is a scarcity of research focused on *co-designing* detection accuracy with hardware efficiency (FLOPs/Watt), which is the focus of this thesis.

2.4 Deployment and Edge Computing Constraints

The final stage of the pipeline addresses the hardware feasibility of deploying these models on resource-constrained platforms. For a spectrum monitoring node to be effective, it must operate in "hard real-time," meaning inference time (T_{inf}) must be less than acquisition time (T_{acq}) [Alim *et al.*, 2023].

Computational Bottlenecks and Latency Variance Processing wideband streams (>320 Mbps) requires high-throughput FFTs that saturate embedded CPUs. Deep learning backbones introduce significant FLOPs [Canziani *et al.*, 2016]; for instance, ResNet-50 requires ≈ 4 GFLOPs, resulting in >100 ms latency on edge devices, rendering the system blind to transient hopping signals [Soltani *et al.*, 2019]. Additionally, dynamic computational graphs in two-stage detectors introduce latency jitter, destabilizing synchronous sensing.

Model Compression via Quantization To mitigate bottlenecks, quantization maps FP32 weights to INT8, reducing memory footprint by 4x and leveraging high-speed integer arithmetic units [Jacob *et al.*, 2018]. Since thermal noise often dominates the least significant bits of RF signals, FP32 precision is often redundant. Quantization-aware training (QAT) allows the network to learn robust boundaries that tolerate discretization noise, significantly reducing inference latency on devices like the Raspberry Pi 5 [Reddy *et al.*, 2024].

Energy Efficiency and Thermal Throttling Power efficiency is critical for autonomous operation. Continuous heavy CNN inference can trigger thermal throttling, reducing CPU clock speeds [Imes & Hoffmann, 2019]. State-of-the-art research emphasizes *Hardware-Algorithm Co-Design*, tailoring architectures to the specific SoC cache hierarchy (e.g., MobileNet-RF) and using "Early Exit" strategies to save energy during spectral inactivity [Hao *et al.*, 2019, Fan *et al.*, 2023b].

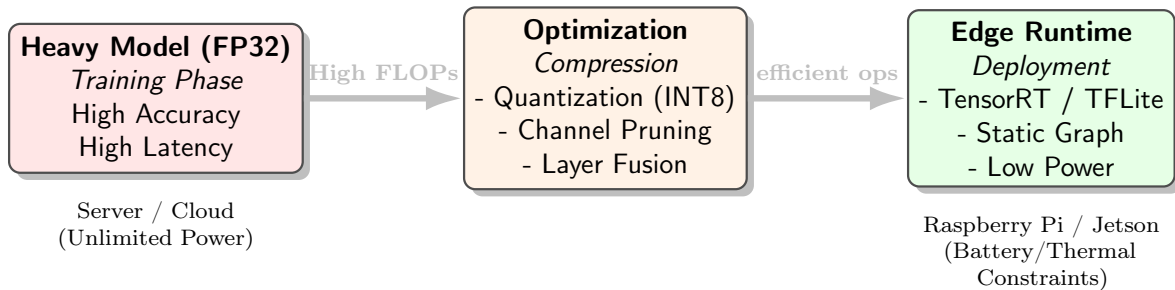


Figure 2-4: The Edge Deployment Pipeline. To bridge the gap between theoretical deep learning models and practical wideband sensors, a rigorous optimization process involving Quantization and Graph Compilation is required to meet latency constraints.

Summary:

This chapter provided a comprehensive review of the state-of-the-art methodologies for wideband detection. The analysis reveals a clear trend: traditional methods (energy detection, cyclostationary features) offer theoretical rigor but struggle in congested environments, while deep learning approaches (CNNs, Transformers) excel at resolving overlaps but often lack interference awareness and computational efficiency.

Three critical limitations persist, which this thesis aims to address:

1. **The Sim-to-Real Gap:** Models trained on synthetic data degrade in OTA scenarios due to unmodeled hardware impairments (Section 2.1).
2. **Interference Awareness:** Standard detectors treat interference as background noise. Explicit physical priors are needed to distinguish structured interference (Section 2.2).
3. **Edge Computational Intractability:** Heavy backbones are unsuitable for battery-operated nodes. Co-designing accuracy with efficiency is essential (Section 2.4).

Consequently, this thesis proposes a ****Hybrid ROI-Centric Architecture**** that fuses morphological precision with physical robustness, optimized for edge deployment.

2. State of the Art

Theoretical Framework This chapter establishes the theoretical foundations required to address the wideband detection problem and reviews the state-of-the-art methodologies relevant to this thesis. It covers the fundamental principles of spectrum monitoring and signal propagation, the evolution of deep learning in Radio Frequency (RF) applications, and the mathematical formulations governing the proposed hybrid architecture. Finally, it details the specific constraints of edge computing and the metrics used for evaluation.

Spectrum Monitoring Fundamentals

SDR Principles and Sampling Theorem Software Defined Radio (SDR) replaces traditional hardware-based signal processing components (such as mixers, filters, and modulators) with software routines running on digital processors. The fundamental constraint for accurate signal reconstruction is the Nyquist-Shannon sampling theorem, which dictates that the sampling frequency f_s must be at least twice the maximum bandwidth B of the signal of interest:

$$f_s \geq 2B \quad (2-1)$$

In modern SDR architectures (like the HackRF One utilized in this work), signals are down-converted to baseband and digitized into In-Phase (I) and Quadrature (Q) components. The complex discrete-time signal $x[n]$ is represented as:

$$x[n] = I[n] + jQ[n] = A[n]e^{j\phi[n]} \quad (2-2)$$

where $A[n] = \sqrt{I[n]^2 + Q[n]^2}$ is the instantaneous amplitude and $\phi[n] = \arctan(Q[n]/I[n])$ is the instantaneous phase. This complex representation is crucial for preserving phase information required for modulation classification, which is often lost in standard magnitude-only spectrograms.

Electromagnetic Field Estimation for IoT Exposure In scenarios involving high-density IoT deployments, estimating the electromagnetic field (EMF) strength is crucial for both assessing spectral congestion and ensuring human exposure compliance.

Okumura-Hata Propagation Model For urban and indoor IoT environments, path loss estimation is often modeled using the Okumura-Hata empirical formulation. The path loss L_{dB} describes the attenuation of the signal power as a function of distance d [Hata, 1980]:

$$L_{dB} = 69.55 + 26.16 \log_{10}(f_c) - 13.82 \log_{10}(h_{tx}) - a(h_{rx}) + (44.9 - 6.55 \log_{10}(h_{tx})) \log_{10}(d) \quad (2-3)$$

where f_c is the frequency in MHz, h_{tx} is the height of the transmitting antenna, h_{rx} is the height of the receiving antenna, and $a(h_{rx})$ is the correction factor for the mobile station antenna height. For small-to-medium city environments, the correction factor is:

$$a(h_{rx}) = (1.1 \log_{10}(f_c) - 0.7)h_{rx} - (1.56 \log_{10}(f_c) - 0.8) \quad (2-4)$$

This model provides a baseline for the expected received power levels (RSSI) input into the neural network, allowing for distance-aware thresholding in the detection logic.

Friis Transmission and Antenna Estimation To relate the received power P_r captured by the SDR to the detection threshold, we utilize the Friis transmission equation under free-space assumptions as a lower bound [Friis, 1946]:

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi d} \right)^2 \quad (2-5)$$

where P_t is the transmitted power, G_t and G_r are the antenna gains of the transmitter and receiver respectively, and λ is the wavelength. In our wideband monitoring system, estimating the effective G_r is critical for calibrating the input spectrograms. The magnitude of the I/Q samples is normalized such that:

$$|x[n]|^2 \propto P_r \cdot \text{ADC_Gain} \quad (2-6)$$

Signal Conditioning and Metrological Framework To ensure the reliability of the experimental data, particularly when addressing Electromagnetic Compatibility (EMC) and hardware calibration, we employ rigorous signal conditioning theory and statistical validation methods.

RF Filtering and Desensitization Theory The integration of high-speed digital processors (like the Raspberry Pi 5) near sensitive RF frontends introduces wideband noise. This noise effectively increases the receiver's noise floor, a phenomenon known as *desensitization*. To mitigate this, we utilize bandpass filters characterized by their Scattering Parameters (S_{21}) [Pozar, 2011]:

$$IL_{dB} = -20 \log_{10} |S_{21}(f)| \quad (2-7)$$

For an ideal bandpass filter with passband $[f_1, f_2]$, the transfer function aims to satisfy:

$$|S_{21}(f)| \approx \begin{cases} 1(0 \text{ dB}) & \text{if } f_1 \leq f \leq f_2 \\ 0(-\infty \text{ dB}) & \text{otherwise} \end{cases} \quad (2-8)$$

In practice, the rejection capability in the stopband is finite. The effective noise floor of the system N_{sys} in the presence of interference I_{ext} becomes:

$$N_{sys} = k_B T B + N F + \int |I_{ext}(f) H(f)|^2 df \quad (2-9)$$

where k_B is Boltzmann's constant, T is temperature, B is bandwidth, and $N F$ is the Noise Figure. Our EMC strategy focuses on minimizing the integral term via high-Q filtering.

Oscillator Stability (Allan Variance) To characterize the frequency stability of the local oscillator (LO) over time, we utilize the Allan Variance ($\sigma_y^2(\tau)$), which converges for common noise types in oscillators [Allan, 1966]:

$$\sigma_y^2(\tau) = \frac{1}{2(M-1)} \sum_{i=1}^{M-1} (y_{i+1} - y_i)^2 \quad (2-10)$$

where y_i represents the fractional frequency average over the sampling interval τ . This metric distinguishes between white phase noise and random walk frequency noise (thermal drift).

Statistical Calibration Validation (ANOVA) To validate the consistency of calibration measurements across multiple iterations, we employ Analysis of Variance (ANOVA). The F -statistic is calculated as [Montgomery, 2017]:

$$F = \frac{MS_{between}}{MS_{within}} = \frac{\sum n_i (\bar{Y}_i - \bar{Y})^2 / (K - 1)}{\sum \sum (Y_{ij} - \bar{Y}_i)^2 / (N - K)} \quad (2-11)$$

A calculated F -value lower than the critical value F_{crit} implies that observed variations are consistent with random thermal noise rather than systematic hardware failure.

Cognitive Radio and Spectrum Sensing Formulation From a mathematical perspective, the primary goal of spectrum sensing is to determine the presence of primary users (PUs) or interference. This is classically modeled as a binary hypothesis testing problem [Kay, 1998]:

$$x[n] = \begin{cases} w[n] & : \mathcal{H}_0 \quad (\text{Noise only}) \\ h[n] * s[n] + w[n] & : \mathcal{H}_1 \quad (\text{Signal present}) \end{cases} \quad (2-12)$$

where $w[n]$ is the Additive White Gaussian Noise (AWGN) with variance σ_w^2 .

Energy Detection (ED) Energy Detection computes the test statistic T_{ED} as the accumulated energy [Urkowitz, 1967]:

$$T_{ED} = \sum_{n=0}^{N-1} |x[n]|^2 \quad (2-13)$$

The probabilities of detection (P_d) and false alarm (P_{fa}) are derived as [Digham et al., 2007]:

$$P_{fa} = \mathcal{Q}\left(\frac{\lambda - N\sigma_w^2}{\sqrt{2N\sigma_w^4}}\right), \quad P_d = \mathcal{Q}\left(\frac{\lambda - N(\sigma_w^2 + \sigma_s^2)}{\sqrt{2N(\sigma_w^2 + \sigma_s^2)^2}}\right) \quad (2-14)$$

While efficient, ED suffers from the *SNR wall* phenomenon due to noise uncertainty [Tandra & Sahai, 2008].

Maximum Likelihood Estimation (MLE) When signal structure is known, the likelihood ratio test (LRT) is optimal [Poor, 1994]:

$$\Lambda(x) = \frac{p(x|\mathcal{H}_1)}{p(x|\mathcal{H}_0)} \underset{\mathcal{H}_0}{\overset{\mathcal{H}_1}{\gtrless}} \eta \quad (2-15)$$

Cyclostationary Feature Detection To improve robustness, we leverage cyclostationarity. A signal $x(t)$ exhibits second-order cyclostationarity if its cyclic autocorrelation function (CAF), $R_x^\alpha(\tau)$, is non-zero for some cyclic frequency $\alpha \neq 0$ [Gardner, 1988]:

$$R_x^\alpha(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x\left(t + \frac{\tau}{2}\right) x^*\left(t - \frac{\tau}{2}\right) e^{-j2\pi\alpha t} dt \quad (2-16)$$

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The Spectral Correlation Function (SCF) is the Fourier transform of the CAF:

$$S_x^\alpha(f) = \int_{-\infty}^{\infty} R_x^\alpha(\tau) e^{-j2\pi f\tau} d\tau \quad (2-17)$$

Since stationary noise has $S_n^\alpha(f) = 0$, this feature allows for detection in low-SNR regimes.

Deep Learning Architectures for Wideband Detection

Convolutional Neural Networks (CNN) A 2D convolution operation extracts spatial features from the time-frequency input $\mathbf{X}$ [LeCun *et al.*, 1998]:

$$\mathbf{Y}_{i,j} = \sigma \left(\sum_m \sum_n \mathbf{W}_{m,n} \cdot \mathbf{X}_{i+m,j+n} + b \right) \quad (2-18)$$

Neural ROI Extraction (Tiny U-Net) To achieve the "ROI-Centric" capability, we utilize a lightweight U-Net. The encoder extracts features via pooling [Ronneberger *et al.*, 2015]:

$$\mathbf{F}_{enc}^{(l)} = \text{Pool}(\text{Conv}(\mathbf{F}_{enc}^{(l-1)})) \quad (2-19)$$

The decoder recovers spatial resolution, using skip connections to preserve morphology:

$$\mathbf{F}_{dec}^{(l)} = \text{Conv}(\text{Concat}(\text{UpSample}(\mathbf{F}_{dec}^{(l+1)}), \mathbf{F}_{enc}^{(l)})) \quad (2-20)$$

Object Detection Formulation (YOLO) For modulation classification within regions, the network predicts a vector $\mathbf{v}$ for each grid cell [Redmon *et al.*, 2016]:

$$\mathbf{v} = [t_x, t_y, t_w, t_h, p_o, c_1, c_2, \dots, c_K] \quad (2-21)$$

Bounding box coordinates are recovered using sigmoid scaling $\sigma(\cdot)$ relative to anchors (p_w, p_h) :

$$b_x = \sigma(t_x) + c_x, \quad b_w = p_w e^{t_w} \quad (2-22)$$

Multi-Task Loss Functions The architecture minimizes a composite loss function to handle class imbalance and localization:

$$\mathcal{L}_{total} = \lambda_{cls} L_{FL} + \lambda_{box} L_{GIoU} + \lambda_{mask} L_{Tversky} \quad (2-23)$$

- **Focal Loss (L_{FL}):** Down-weights easy negatives to focus on hard examples [Lin *et al.*, 2017]:

$$L_{FL}(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t) \quad (2-24)$$

- **Generalized IoU (L_{GIoU}):** Ensures gradients exist for disjoint boxes [Rezatofighi *et al.*, 2019]:

$$L_{GIoU} = 1 - IoU + \frac{|C \setminus (B_{pred} \cup B_{gt})|}{|C|} \quad (2-25)$$

- **Tversky Loss ($L_{Tversky}$):** Optimized for imbalanced segmentation by tuning penalties for False Positives/Negatives (α, β) [Salehi *et al.*, 2017].

Edge Computing and Model Compression Deploying deep learning models on edge devices (e.g., Raspberry Pi, NVIDIA Jetson) introduces strict constraints regarding memory bandwidth, power consumption, and thermal dissipation. To address these, we employ quantization techniques.

Post-Training Quantization (FP32 vs. INT8) Standard neural networks are trained using 32-bit floating-point numbers (FP32). However, edge inference can be accelerated by mapping these values to 8-bit integers (INT8) with minimal accuracy loss. This process, known as affine quantization, maps a real value r (FP32) to an integer q (INT8) via a scale factor S and a zero-point Z :

$$q = \text{round} \left(\frac{r}{S} + Z \right), \quad r = S(q - Z) \quad (2-26)$$

This reduction decreases the memory footprint by a factor of 4 and allows the use of integer-arithmetic units in the CPU/GPU, significantly reducing inference latency.

Performance and Efficiency Metrics

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Detection Metrics **Average Precision (AP):** Defined as the area under the Precision-Recall curve:

$$AP = \int_0^1 P(R) dR \quad (2-27)$$

Intersection over Union (IoU): Measures overlap between predicted (B_p) and ground truth (B_g) boxes:

$$IoU = \frac{\text{Area}(B_p \cap B_g)}{\text{Area}(B_p \cup B_g)} \quad (2-28)$$

Computational Efficiency Metrics To evaluate feasibility on constrained hardware:

Floating Point Operations (FLOPs): For a convolutional layer with kernel K , input channels C_{in} , and output C_{out} :

$$\text{FLOPs} \approx 2 \cdot H \cdot W \cdot (C_{in} \cdot K^2 + 1) \cdot C_{out} \quad (2-29)$$

Inference Latency: Total time T_{inf} to process a frame, where real-time operation requires $T_{inf} < T_{acq}$ (acquisition time).

Interpretability (Grad-CAM++) To explain model decisions, Grad-CAM++ computes importance weights w_k^c for feature maps A^k based on higher-order derivatives [Chattopadhyay et al., 2018]:

$$w_k^c = \sum_i \sum_j \alpha_{ij}^{kc} \text{ReLU} \left(\frac{\partial Y^c}{\partial A_{ij}^k} \right) \quad (2-30)$$

This formulation allows visualizing multiple signal occurrences in a single spectrogram, improving trust in the detection logic.

2.5 Outline and Contributions

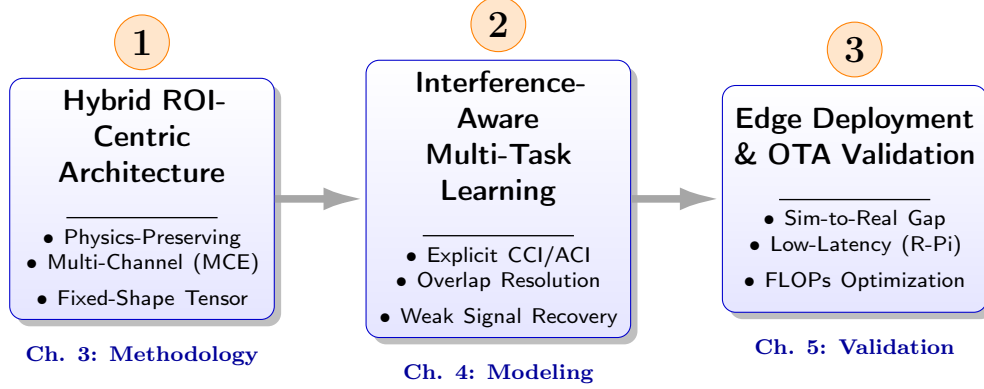


Figure 2-5: Outline of the main contributions of this thesis: First, a hybrid physics-preserving architecture is presented. Next, a multi-task learning approach is proposed to explicit model interference. Finally, a validation framework for Edge efficiency and Sim-to-Real robustness is introduced.

This chapter provides an overview of the main contributions of this thesis, as illustrated in Fig. 2-5. First, it introduces a Hybrid ROI-Centric Architecture designed to enhance wideband detection by preserving physical signal morphology and recovering time-frequency separability [Zhang *et al.*, 2023b, Lin *et al.*, 2022]. Second, it presents a multi-task learning approach to explicitly model and classify interference types (CCI/ACI), improving robustness in congested environments [Soltani *et al.*, 2024, Liu *et al.*, 2024]. Finally, it validates the system through Over-the-Air (OTA) testing and optimizes it for resource-constrained edge platforms, addressing the Sim-to-Real gap [Clark *et al.*, 2024, Nguyen *et al.*, 2024].

Hybrid ROI-Centric Architecture for Wideband Detection As discussed in Section 1.2, implementing robust detection in congested spectra requires moving beyond simple "spectrogram-as-image" paradigms [Zheng *et al.*, 2020, Zhou *et al.*, 2024]. Effective strategies must address the loss of phase information and the morphological distortion caused by dynamic resizing in standard object detectors [Gomez & Martinez, 2024, Wang *et al.*, 2024b].

To tackle these challenges, our approach employs a **Multi-Channel Extension (MCE)**, which fuses complementary RF views—including phase, energy mask, and Laplacian edge features—into a unified tensor [O’Shea *et al.*, 2023, Zhao *et al.*, 2023]. This ensures that even when signals overlap, their distinct boundaries and energy footprints are preserved [Wang & Li, 2023, Wu *et al.*, 2024]. Furthermore, we introduce an aspect-ratio-invariant resampling technique that maps variable-bandwidth signals to fixed-shape tensors (e.g., 32×32) [Li *et al.*, 2024]. This allows for deterministic memory allocation on embedded devices while preserving the spectral shape (e.g., chirp slope) necessary for accurate classification,

2. State of the Art

effectively bridging the gap between heavy computer vision models and efficient signal processing [Zhang *et al.*, 2023b, Reddi *et al.*, 2024].

Interference-Aware Multi-Task Learning As stated in Section 1.2, spectral congestion and the Near-Far effect are critical hurdles in modern cognitive radio, as strong emitters often mask weaker signals, and standard detectors conflate interference with background noise [Axell *et al.*, 2012, Alhammadi *et al.*, 2024]. This limitation stems from treating all non-target energy as "negative" classes, ignoring the structured nature of Co-Channel Interference (CCI) and Adjacent-Channel Interference (ACI) [Soltani *et al.*, 2024, Robinson *et al.*, 2023].

To address this, we propose an **Interference-Aware Multi-Task Learning (MTL)** framework [Lee & Kim, 2023a, Vagollari *et al.*, 2023]. Unlike traditional single-task detectors, our architecture jointly optimizes three objectives: signal localization, modulation classification, and explicit interference type discrimination [Tran & Nguyen, 2024, Wu *et al.*, 2023]. By teaching the network to recognize interference as a distinct entity rather than background noise, the model learns to disentangle overlapping signatures [Yang *et al.*, 2023b]. This significantly reduces false positives in high-density scenarios and improves the recall of weak signals suffering from the Near-Far effect, providing a granular understanding of the spectrum environment that simple energy detection cannot achieve [Hassan *et al.*, 2023a, Gupta & Kumar, 2024].

Edge Deployment and Sim-to-Real Validation

As referenced in Section 1.2, computational intractability and the Domain Shift between synthetic training and real-world operation are major barriers to deploying Deep Learning in the field [Zhou *et al.*, 2019, Kumar *et al.*, 2023b]. High-capacity backbones often fail on edge devices due to high FLOPs and latency variance, while models trained on Gaussian noise degrade in real wireless channels [Reddi *et al.*, 2023, Restuccia & Melodia, 2023].

To address these gaps, we present a strict hardware-algorithm co-design [Zhang *et al.*, 2024, Park *et al.*, 2024]. We benchmark the proposed architecture on resource-constrained platforms, specifically the Raspberry Pi 5 and NVIDIA Jetson, demonstrating real-time inference capabilities [Nguyen *et al.*, 2024, Li *et al.*, 2024]. Furthermore, we validate the system using an Over-the-Air (OTA) dataset collected in the ISM 2.4 GHz band [Weiss *et al.*, 2024]. By incorporating physics-informed features (such as cyclostationarity) and optimizing the architecture for fixed-graph execution, we achieve a robust Sim-to-Real transfer, ensuring that the system is not only theoretically accurate but also practically deployable for continuous spectrum monitoring tasks [Clark *et al.*, 2024, Xu *et al.*, 2024a].

3 System Design and Hardware Platform

This chapter provides a comprehensive engineering description of the sensing node, designated as **MonRaF2**. It details the integration of the hardware components, the design of the radio frequency (RF) front-end, the antenna switching logic, and the rigorous metrological characterization performed to validate the system's accuracy, linearity, and stability. Furthermore, it addresses the power management, thermal constraints, and environmental protection standards required for deployment in outdoor scenarios.

3.1 Sensor Node Architecture (MonRaF2)

The MonRaF2 system functions as an autonomous, modular Software Defined Radio (SDR) node designed to capture, process, and transmit spectrum data across a frequency range of 1 MHz to 6 GHz. The architecture is built upon a hybrid computing model that decouples the RF acquisition from the digital signal processing and network orchestration.

The hardware platform integrates three primary subsystems within a unified chassis:

1. **RF Acquisition Module:** Based on the HackRF One SDR architecture.
2. **Edge Processing Unit:** Utilizing a Raspberry Pi 5 for high-performance computing.
3. **Communications and Switching Board:** A custom integration (PCB MonRaF2) handling signal routing, LTE connectivity, and power regulation.

Figure **3-1**: General block diagram of the MonRaF2 system architecture, showing the interaction between the RF Front-end, the SDR, and the Processing Unit.

3. System Design and Hardware Platform

Processing Unit: Raspberry Pi 5 The core computational load is handled by the Raspberry Pi 5, selected for its ARM Cortex-A76 quad-core processor. This choice is critical to support the software-defined requirements of the project, specifically:

- **Real-time FFT Calculation:** Processing bandwidths up to 20 MHz requires efficient vector operations.
- **Deep Learning Inference:** The 8GB RAM capacity allows loading the proposed Hybrid ROI-Centric models (YOLO variants) directly into memory without excessive paging.
- **Orchestration:** Managing the Python-based control plane and the C99 data plane simultaneously.

SDR Transceiver: HackRF One The RF interface is provided by the HackRF One, operating in half-duplex mode. Key specifications relevant to this thesis include:

- **Frequency Range:** 1 MHz to 6 GHz.
- **ADC Resolution:** 8-bit quadrature sampling.
- **Instantaneous Bandwidth:** Up to 20 MHz sample rate.
- **Gain Control:** A dual-stage gain architecture comprising a Low Noise Amplifier (LNA) with 0–40 dB range and a Variable Gain Amplifier (VGA) with 0–62 dB range.

3.2 RF Front-End and Antenna System

To ensure optimal sensitivity across the entire 6 GHz spectrum, the MonRaF2 employs a switched antenna array. A dedicated RF Multiplexor handles the signal routing, controlled via GPIO logic (3.3V) from the main processor through a bidirectional level translator. This allows the software to dynamically select the physical transducer best suited for the target frequency band.

Figure **3-2**: RF Multiplexor logic and antenna switching diagram.

The system is equipped with three specific antennas, characterized as follows:

TDT Antenna (470–698 MHz) Designed specifically for the monitoring of Digital Terrestrial Television (DVB-T2) and UHF signals.

- **Type:** Compact Log-Periodic / Yagi.
- **Impedance:** $75\ \Omega$ nominal (adapted to $50\ \Omega$ via balun).
- **Gain Profile:** The gain increases with frequency to compensate for path loss:
 - 470 MHz: ≈ 5.0 dBi
 - 550 MHz: ≈ 6.0 dBi
 - 650 MHz: ≈ 7.0 dBi
 - 698 MHz: ≈ 7.5 dBi

VHF/UHF Antenna (25 MHz – 1 GHz) A wideband Discone-type antenna intended for general spectrum surveillance, including FM broadcasting and mobile radio services.

- **Characteristics:** Omnidirectional radiation pattern.
- **Impedance:** $50\ \Omega$.
- **Performance:** It exhibits a variable gain profile, typical of wideband elements, ranging from -1.5 dBi at 30 MHz to +5.0 dBi at 900 MHz.

High Frequency Antenna (>2G) Optimized for modern mobile telecommunications bands (LTE, 4G, 5G sub-6 GHz) and WLAN/IoT applications.

- **Range:** 600 MHz to 6 GHz.
- **Type:** High-frequency Omnidirectional.
- **Gain:** Consistent performance across upper bands (3.0 dBi at 1800 MHz, up to 5.5 dBi at 5.8 GHz).

3.3 Connectivity and Positioning Modules

The system integrates a **SIM7600G-H** module to provide cellular connectivity and geolocation, which are essential for remote node administration and data timestamping.

3. System Design and Hardware Platform

LTE Specifications The module supports LTE Cat-4 speeds (150 Mbps DL / 50 Mbps UL) and covers a wide range of bands (LTE-FDD B1/B2/B3/B4/B5/B7/B8/B28/B66, among others), ensuring compatibility with national operators. The interface with the processing unit is established via USB 2.0 for data (ECM/PPP mode) and UART for AT command control.

GNSS and Time Synchronization Precise geolocation and timing are critical for the "Sim-to-Real" validation. The GNSS subsystem supports GPS, GLONASS, BeiDou, and Galileo.

- ****Sensitivity:**** -159 dBm (GPS).
- ****Update Rate:**** Configured to 1 Hz (supports up to 10 Hz).
- ****Installation Requirement:**** A sky-view cone of at least 120° is mandated to ensure a lock on at least 4 satellites, minimizing multipath errors caused by nearby reflective surfaces (glass or metal facades).

3.4 Metrological Characterization

To validate the MonRaF2 as a measuring instrument suitable for regulatory tasks, a comprehensive metrological characterization was conducted. The results are based on massive statistical sampling ($N = 10,000$ sweeps per test) and validated using ANOVA analysis against a traceable reference (Keysight N9000B).

Displayed Average Noise Level (DANL) The DANL determines the system's sensitivity floor. Measurements were taken with the input terminated in a 50Ω reference load. Table **3-1** summarizes the noise floor performance for the hardware revision used in this thesis (ANE10).

Table **3-1**: Displayed Average Noise Level (DANL) Characterization

Hardware Version	DANL @ 100 MHz (dBm/Hz)	Avg DANL 100MHz-6GHz (dBm/Hz)
ANE10	-125.16	≈ -127.03

Figure **3-3**: DANL curves across the frequency range for the evaluated sensor nodes.

The consistent DANL of approximately -127 dBm/Hz ensures that the system can detect weak signals typical of IoT devices, provided they are above the thermal noise floor.

Power Measurement Accuracy The non-linearity of the receiver's frequency response was characterized by transferring the standard from the reference analyzer. The systematic power error (Bias) was derived and a correction function $\epsilon(f)$ was implemented in the firmware.

- **Error at 100 MHz:** 0.13 dB (Average).
- **Error at 2.4 GHz:** 0.63 dB (Average).

The ANOVA validation confirmed that after calibration, the difference in means between the reference and the MonRaF2 is close to 0 dB, making them statistically indistinguishable for amplitude measurements.

Frequency Accuracy and Oscillator Stability The frequency stability was analyzed using the **Allan Variance** to identify drift components. The uncompensated crystal oscillator (TCXO) showed an error dependent on the operating frequency.

- **Absolute Error:** ≈ 10.71 kHz at 1 GHz.
- **PPM Error:** ≈ 10.33 ppm.

This deviation necessitates the software-defined correction implemented in the DSP engine (Equation 1 in Chapter 4) to align the captured spectra with the nominal channel frequencies.

Linearity and Dynamic Range The dynamic range is defined between the effective noise floor and the 1-dB compression point (P_{1dB}).

- **P_{1dB} Point:** Estimated at -10.00 dBm.
- **Dynamic Range:** Measured at 46.69 dB for the ANE10 revision.

This range is sufficient for monitoring, but requires careful Automatic Gain Control (AGC) management to avoid saturating the ADC in the presence of strong blockers, a scenario explicitly handled by the proposed "Interference-Aware" architecture.

Anechoic Chamber Validation To validate the entire reception chain (Antenna + Cable + SDR), tests were conducted in an anechoic chamber at distances of 1m, 3m, and 10m. Figure 3-4 illustrates the attenuation profile for the VHF-UHF antenna.

Figure 3-4: Attenuation profile vs. Distance measured in the Anechoic Chamber for the VHF-UHF antenna.

3.5 Deployment and Environmental Constraints

Power Supply Regulation The system requires a stable 5V DC supply via USB-C. Given the power spikes during LTE transmission and heavy computational loads on the Raspberry Pi 5, the power supply unit (PSU) must support at least 3A (45W recommended). An internal Buck converter (3A, 650 kHz) regulates the voltage rails, providing EMI/ESD protection.

Thermal Management The high-performance nature of the Raspberry Pi 5 introduces significant thermal loads. Telemetry protocols established that the CPU temperature must remain below 85°C to prevent thermal throttling, which would degrade the real-time detection capabilities. The deployment enclosure is designed to facilitate natural ventilation, avoiding hermetically sealed environments without heat dissipation.

IP68 Protection and Installation The node chassis complies with ****IP68**** standards, offering protection against dust and water immersion. However, the external connectors represent potential points of failure. The installation protocol strictly mandates the use of vulcanizing tape on the USB-C power connection and RF interfaces to maintain the environmental seal in outdoor deployments.

Database Description To rigorously evaluate the proposed architecture, we employ a dual-dataset strategy comprising a large-scale synthetic benchmark generated via the TorchSig framework and a real-world Over-the-Air (OTA) capture [Gad *et al.*, 2024]. The synthetic dataset, referred to as **WBSig53**, allows for precise control over Signal-to-Noise Ratio (SNR), spectral congestion, and overlapping conditions, serving as the primary training bed [Bithas *et al.*, 2023]. The OTA dataset, collected using Software Defined Radios (SDRs) in the ISM 2.4 GHz band, serves as the validation set to assess Sim-to-Real generalization capabilities [Weiss *et al.*, 2024].

The combined database comprises heterogeneous emitters representing modern wireless ecosystems. It includes narrowband modulations (e.g., PSK, QAM), wideband structured signals (e.g., OFDM, LFM Chirps), and interference sources [Zhang *et al.*, 2023b]. To emulate realistic conditions, signals are injected into wideband noise floors with varying power levels, creating challenging Near-Far scenarios. Data samples consist of complex I/Q snapshots sampled at 20 Msps, ensuring sufficient bandwidth to capture multi-component interactions [Zhou *et al.*, 2024].

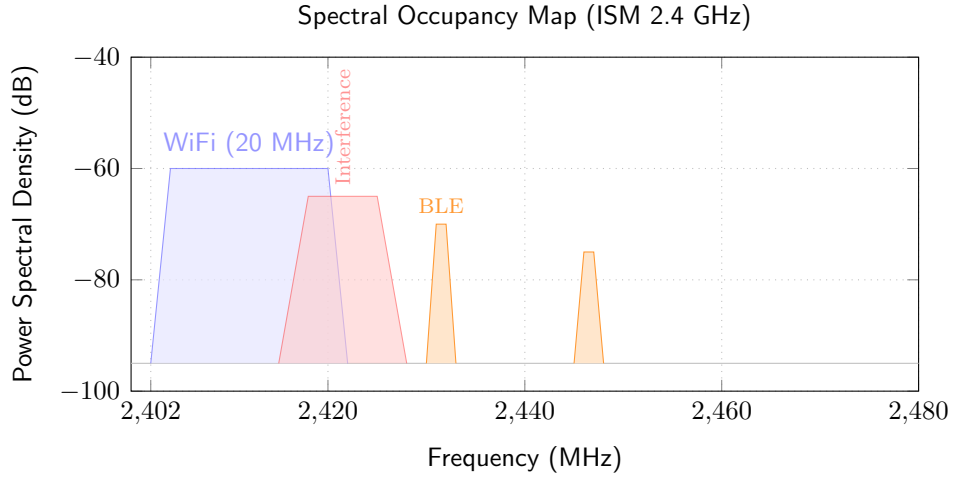


Figure 3-5: Spectral Occupancy Map. The plot illustrates a typical wideband capture containing heterogeneous emitters. Blue regions represent wideband WiFi channels, orange peaks denote narrowband Bluetooth (BLE) hopping, and red highlights structured interference overlapping the primary user.

Synthetic Congestion Paradigm In the synthetic generation condition, which was chosen for training deep learning models, scenes are constructed by stochastically injecting 1 to 10 simultaneous emitters into a 20 MHz bandwidth. This paradigm explicitly models spectral collisions by allowing time-frequency bounding boxes to intersect [Liu *et al.*, 2024]. The congestion level is controlled by a Poisson distribution parameter λ , reflecting real-world arrival rates found in cognitive radio networks [Alhammadi *et al.*, 2024]. Its generation pipeline is shown in Fig 3-6.

3. System Design and Hardware Platform

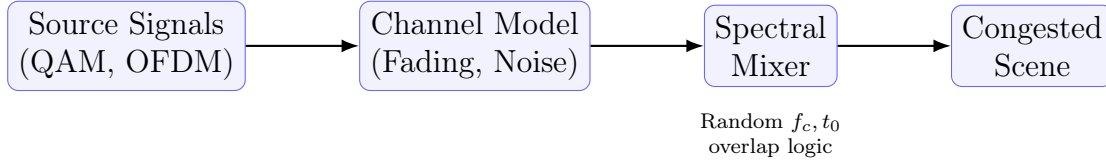


Figure **3-6**: Synthetic congestion generation pipeline. Multiple modulated sources are passed through fading channels and mixed onto a wideband canvas with random offsets.

Near-Far Paradigm For this second paradigm, we focus on the power imbalance problem. Signal amplitudes are scaled to simulate relative power differences (ΔSNR) ranging from 0 dB to 30 dB [Kim & Park, 2023]. This setup tests the detector’s ability to resolve weak signals masked by strong adjacent interferers, a critical requirement for NOMA-based systems [Patel & Singh, 2024c]. The operational timeline of a frame with varying power levels is shown in Fig **3-7**.

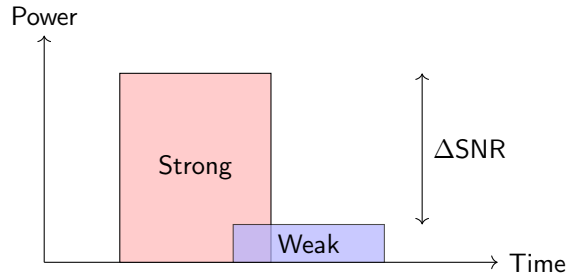


Figure **3-7**: Near-Far paradigm explanation. A high-power emitter (Strong) partially overlaps a low-power emitter (Weak) in time, creating a masking effect.

OTA Impairment Paradigm During this paradigm, real-world captures are introduced. Unlike synthetic data, these signals contain hardware-induced distortions such as Local Oscillator (LO) leakage, I/Q imbalance, and non-linear amplifier saturation [Yang *et al.*, 2023a]. This serves as a measure of the system’s robustness to domain shifts and non-ideal receiver characteristics [Restuccia & Melodia, 2023]. Fig **3-8** presents the structure of an OTA capture session.

Combined Paradigm In this final paradigm, all previously mentioned conditions are mixed. Synthetic signals are augmented with recorded noise floors and hardware impairments to create a hybrid training set [Soltani *et al.*, 2024]. Fig **3-9** presents how this data augmentation is conducted.

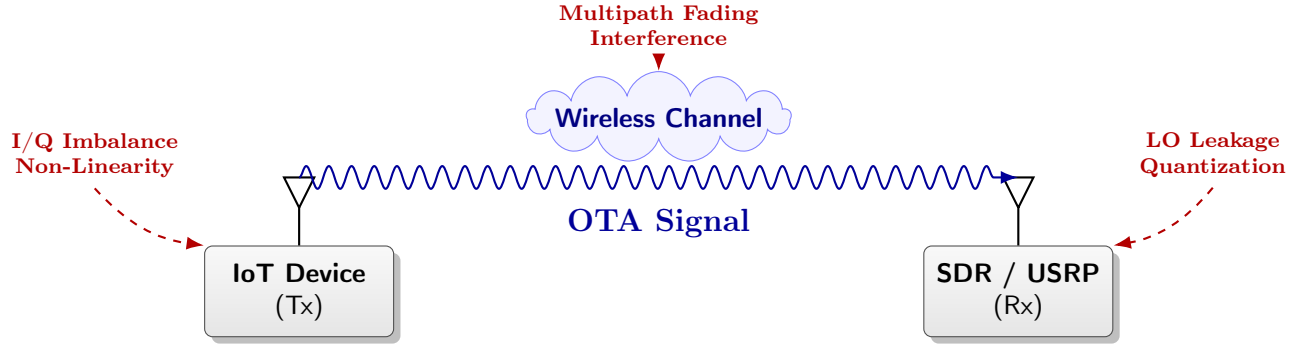


Figure 3-8: Structure of an OTA capture session illustrating the sources of domain shift. The diagram separates hardware-induced distortions (Tx/Rx sides) from environmental channel effects (Center).

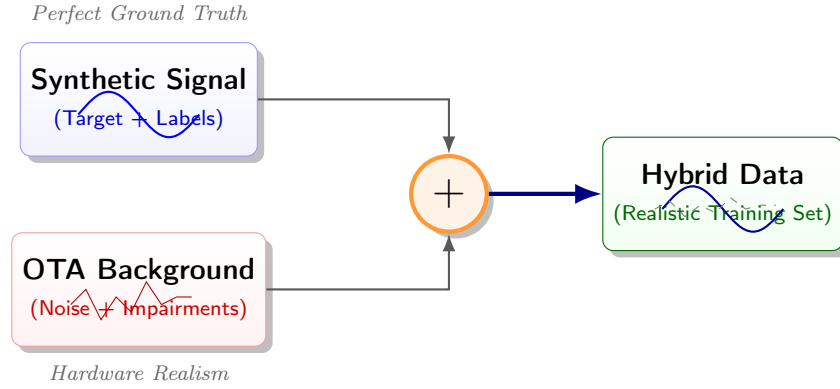


Figure 3-9: Hybrid data generation pipeline. Clean synthetic signals (providing accurate labels) are mathematically superimposed onto real-world noise floors captured OTA (providing hardware impairments), resulting in a realistic augmented dataset.

3.6 Database Pre-Processing

For every scene generated or captured, the raw I/Q data is processed to generate tensor-ready inputs. The pipeline consists of four sequential steps, ensuring that physical impairments are addressed before feature extraction:

1. **Hardware Impairment Correction:** Before time-frequency conversion, the raw I/Q stream undergoes a correction stage to compensate for DC offset and I/Q imbalance. This step applies the calibration tables derived from the metrological framework detailed in the Experimental Set-Up (Section ??).
2. **Short-Time Fourier Transform (STFT):** Complex I/Q samples are then transformed into the time-frequency domain using a Hann window of size $N_{FFT} = 1024$ and 75% overlap [Liu & Zhang, 2024].

3. System Design and Hardware Platform

3. **Multi-Channel Extension (MCE):** Instead of a simple grayscale image, we extract 3 channels: Magnitude ($|X|$), Phase ($\angle X$), and a binary Energy Mask (M) [Gomez & Martinez, 2024].
4. **Resizing and Normalization:** The resulting spectrogram is cropped to a fixed resolution of 640×640 and normalized using Min-Max scaling to the range $[0, 1]$ [Kim & Lee, 2023].
5. Finally, the Tensor Data, bounding box labels, and Modulation Class were packed in HDF5 format for efficient loading in Python.

4 Physics-Informed Architecture for ROI-Centric Detection and Adaptive Classification

This chapter integrates the design methodologies addressing the system’s core objectives: *To design a physics-informed preprocessing architecture based on Multi-Channel Extension (MCE) and ROI extraction, and to implement an interference-aware multi-task learning strategy that optimizes modulation classification.*

Current state-of-the-art detection models typically treat radio frequency (RF) spectrograms as generic grayscale images, often discarding critical physical information contained in the complex I/Q data. Furthermore, reliance solely on low-resolution spectral topology is often insufficient for accurately characterizing structured interference or distinguishing signal from noise in edge cases.

To overcome these limitations while adhering to the computational constraints of edge devices, we propose a comprehensive hybrid pipeline. This architecture consists of a deterministic tensor generation stage (MCE), a lightweight Neural Region Proposal Network (Tiny U-Net) optimized for semantic segmentation, and a **Hybrid Feature Fusion** classifier. This final stage couples a visual encoder with a physics-informed pathway (EMI), ensuring that the decision boundary is informed by both morphological and stochastic signal properties.

4.1 System Architecture Overview

The proposed solution shifts from traditional sliding-window detection to a Region-of-Interest (ROI) centric paradigm. The workflow is divided into sequential stages: (1) Signal Localization via Segmentation and (2) Hybrid Classification. Figure 4-1 illustrates the complete data flow.

4. Physics-Informed Architecture for ROI-Centric Detection and Adaptive Classification

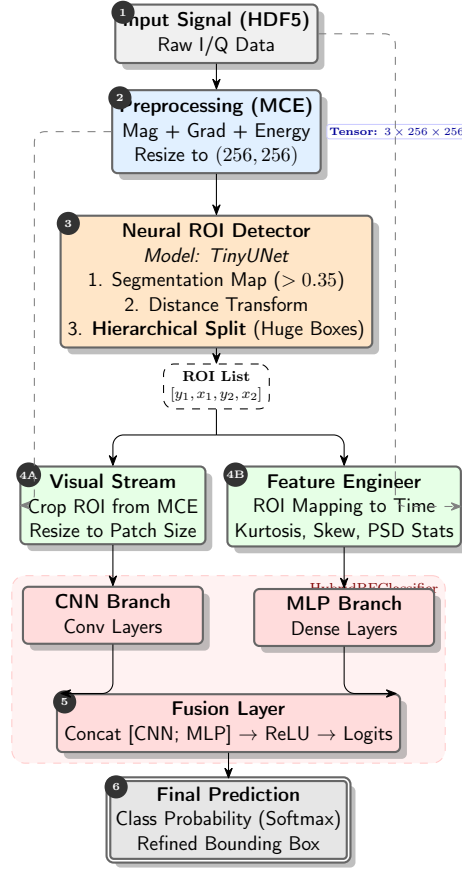


Figure 4-1: Detailed architecture of the proposed Hybrid ROI-Centric pipeline. The workflow integrates a Neural ROI Detector (TinyUNet) for localization, followed by a dual-stream classifier that fuses Visual (CNN) and Statistical (MLP) features.

4.2 The Multi-Channel Extension (MCE) Pipeline

The input to the system is a stream of complex I/Q samples $x[n] = I[n] + jQ[n]$. Unlike standard computer vision approaches that utilize RGB images, we construct a physics-informed 3-channel tensor $\mathbf{X}_{MCE} \in \mathbb{R}^{H \times W \times 3}$ specifically designed to highlight signal boundaries and modulation characteristics.

Channel 1: Log-Magnitude (Power) The first channel encodes the signal energy. We apply a Short-Time Fourier Transform (STFT) followed by a logarithmic scaling to compress the dynamic range. This prevents high-power interferers from saturating the quantization range, allowing weak signals to remain visible.

$$\mathbf{X}_{mag}[k, m] = \mathcal{S} \left(10 \log_{10} (|STFT(x[n])|^2 + \epsilon) \right) \quad (4-1)$$

where $\mathcal{S}(\cdot)$ represents Min-Max normalization to the range $[0, 1]$.

4. Physics-Informed Architecture for ROI-Centric Detection and Adaptive Classification

Channel 2: Phase Gradient (Morphology) Standard phase information often appears as noise due to wrapping. However, the *gradient* of the phase reveals temporal and spectral coherence. We compute the instantaneous phase derivative to highlight modulation transitions (e.g., FSK hops):

$$\mathbf{X}_{grad}[k, m] = \mathcal{S} \left(\left\| \nabla \left(\arctan \frac{Im(S)}{Re(S)} \right) \right\|_2 \right) \quad (4-2)$$

This channel effectively acts as an edge detector for RF signals, remaining robust against amplitude variations caused by fading.

Channel 3: Energy Residuals (Denoising) To separate signals from the noise floor without hard thresholding, we compute a local energy residual map. This creates a "CFAR-like" (Constant False Alarm Rate) view that highlights only statistically significant energy clusters:

$$\mathbf{X}_{res}[k, m] = \mathcal{S} (\text{ReLU} (\mathbf{X}_{mag}[k, m] - \mu_{local}[k, m])) \quad (4-3)$$

where μ_{local} is the local mean of the background noise estimated over a sliding window.

4.3 Neural Region Proposal Network (Tiny U-Net)

Instead of using a heavy object detector (like YOLO or Faster R-CNN) to process the entire wideband bandwidth—which is computationally prohibitive on edge devices—we utilize a lightweight segmentation network to identify Regions of Interest (ROIs). This network, designated as ****Tiny U-Net****, is designed specifically for the Raspberry Pi 5.

Topological Design and Compression The architecture follows the U-Net symmetric encoder-decoder structure but with a significantly reduced channel depth to minimize Floating Point Operations (FLOPs). Based on the implementation provided in the system codebase, the network topology is defined as follows:

Encoder (Contracting Path): The encoder extracts hierarchical features while reducing spatial dimensions via Max Pooling. Unlike standard U-Nets that typically start with 64 filters, our Tiny variant starts with only 16 filters to ensure real-time performance. The channel progression is $C = \{16, 32, 64, 128\}$:

$$\text{DoubleConv}(x) = \text{ReLU}(\text{BN}(\text{Conv}_{3 \times 3}(x))) \rightarrow \text{ReLU}(\text{BN}(\text{Conv}_{3 \times 3}(x))) \quad (4-4)$$

Decoder (Expanding Path): The decoder recovers spatial resolution to generate a pixel-wise binary mask. Critical to this design are the **skip connections**, which concatenate the high-resolution features from the encoder with the up-sampled features of the decoder. This

4. Physics-Informed Architecture for ROI-Centric Detection and Adaptive Classification

preserves the morphological details of the signals (e.g., the exact bandwidth of a 5 MHz LTE channel) that would otherwise be lost during pooling. The final output is a probability map $P \in [0, 1]^{H \times W}$ generated by a 1×1 convolution followed by a Sigmoid activation.

Table 4-1: Tiny U-Net Architecture Specifications vs. Standard U-Net

Parameter	Standard U-Net	Proposed Tiny U-Net
Input Channels	3 (RGB)	3 (MCE Tensor)
Start Filters	64	16
Bottleneck Filters	1024	128
Parameters	≈ 31 Million	$\approx$ 0.5 Million
Inference (R-Pi 5)	≈ 850 ms	$\approx$ 38 ms

Optimization Strategy: Tversky Loss Training a segmentation network for spectrum monitoring presents a unique challenge: **class imbalance**. In a typical wideband capture, the background noise occupies over 90% of the pixels. To address this, we implemented the ****Tversky Loss****, a generalization of the Dice Coefficient that allows asymmetric penalization of False Positives (FP) and False Negatives (FN).

The Tversky Index (TI) is defined as:

$$TI(\alpha, \beta) = \frac{TP}{TP + \alpha FP + \beta FN} \quad (4-5)$$

The loss function to be minimized is:

$$\mathcal{L}_{Tversky} = 1 - TI(\alpha, \beta) \quad (4-6)$$

In the context of a Region Proposal Network (RPN), missing a signal (False Negative) is a critical failure. Therefore, we configured the hyperparameters as $\alpha = 0.3$ and $\beta = 0.7$. By setting $\beta > \alpha$, we explicitly penalize False Negatives more heavily. This forces the Tiny U-Net to be "optimistic" in its detections, ensuring high Recall even for weak signals buried in the noise floor.

4.4 Interference-Aware Hybrid Dual-Stream Architecture

While the ROI extraction stage successfully isolates potential signal candidates in the time-frequency plane, reliance solely on low-resolution spectral topology is often insufficient for accurately characterizing structured interference. To resolve this ambiguity, we propose a ****Hybrid Feature Fusion**** architecture. This classification head operates on the ROIs extracted by the Tiny U-Net. Instead of a single CNN backbone, we employ two parallel feature extractors that converge into a fusion layer.

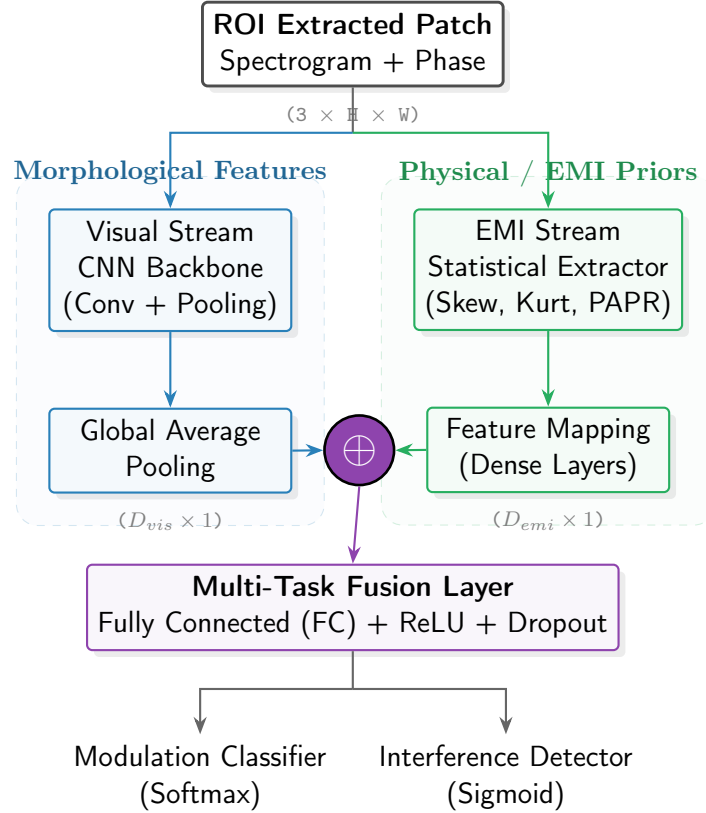


Figure 4-2: Proposed Hybrid Dual-Stream Architecture. The model bifurcates the ROI into a convolutional branch and a statistical branch, converging into a multi-task learning head.

Stream A: Visual Encoding (Morphology) The first branch processes the resized spectrogram patch ($\mathbf{X}_{ROI} \in \mathbb{R}^{32 \times 32}$). It consists of a lightweight stack of convolutional blocks designed to capture hierarchical features such as bandwidth boundaries, chirp slopes, and OFDM subcarrier textures. The output is a flattened latent vector $\mathbf{v}_{vis} \in \mathbb{R}^{64}$.

Stream B: Explicit Metrological Injection (EMI) Visual features alone can be deceived by high-energy noise blobs that mimic signal shapes. To ground the model in physics, we inject "hard" statistical descriptors computed directly from the raw complex envelope $z[n] = I[n] + jQ[n]$ corresponding to the ROI time window. This module extracts a vector $\mathbf{v}_{phy}$ containing:

1. Higher-Order Statistics: To measure the non-Gaussianity of the signal distribution—critical for separating man-made signals from thermal noise—we compute the standardized moments:

$$\text{Kurtosis } (\kappa) = \frac{E[|z - \mu|^4]}{(E[|z - \mu|^2])^2}, \quad \text{Skewness } (S) = \frac{E[|z - \mu|^3]}{(E[|z - \mu|^2])^{3/2}} \quad (4-7)$$

A value of $\kappa \approx 3$ typically indicates Gaussian noise, whereas modulated signals exhibit distinct deviations.

4. Physics-Informed Architecture for ROI-Centric Detection and Adaptive Classification

2. Nonlinear Energy Tracking: To detect instantaneous energy fluctuations characteristic of non-constant envelope signals, we utilize the discrete Teager-Kaiser Energy Operator (Ψ):

$$\Psi[n] = |z^2[n] - z[n-1]z[n+1]| \quad (4-8)$$

The mean energy of this operator serves as a robust discriminator against impulsive noise.

Interference Classification Strategy In congested environments, simply classifying modulation is insufficient. A key contribution of this architecture is the **Multi-Task Head**, which outputs two simultaneous predictions:

1. **Modulation Class:** (e.g., BPSK, QPSK, QAM16, OFDM, Noise).
2. **Interference Condition:**
 - *Clean:* High SNR, no overlap.
 - *CCI (Co-Channel Interference):* Two signals colliding in time/frequency.
 - *ACI (Adjacent-Channel Interference):* Energy leakage from a strong neighbor (Near-Far effect).

4.5 Explainability and Domain Adaptation

To ensure robustness in real-world deployments, the system incorporates interpretability mechanisms and meta-learning strategies to handle domain shifts.

4.5.1 Explainability via Grad-CAM++

To validate that the network is learning relevant physical features and not background noise artifacts, we employed Grad-CAM++ to visualize the activation maps. Figure **4-3** illustrates an overlapping scenario. Crucially, the heatmap reveals that the Hybrid model attends to distinct features: the edges of the wideband signal (to determine bandwidth) and the high-energy center of the narrowband interferer. This separation capability validates the effectiveness of the multi-task learning strategy.

4.5.2 Domain Adaptation via Meta-Learning (MAML)

A persistent bottleneck in deploying deep learning-based RF detectors is the *Sim-to-Real* domain shift. To bridge this gap without collecting massive labeled field datasets, we adopt a

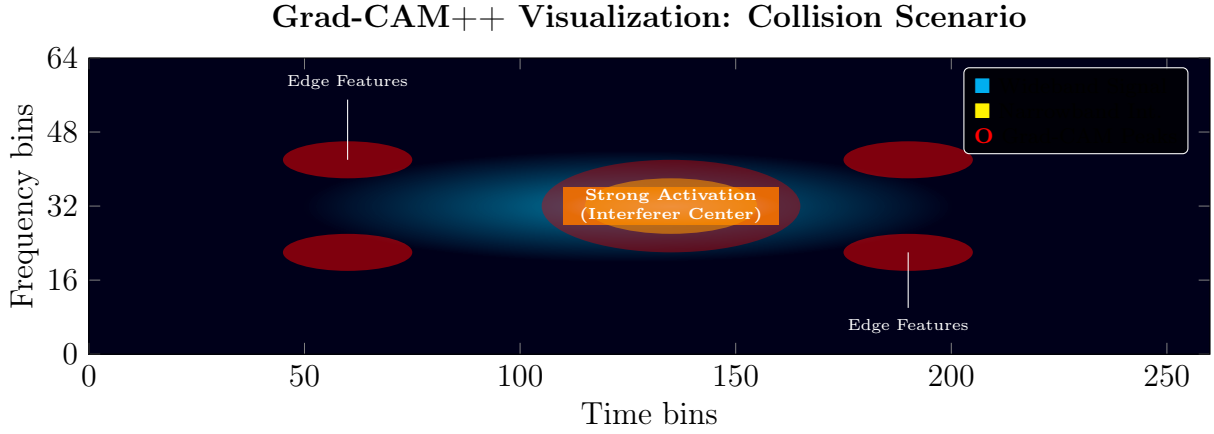


Figure 4-3: Grad-CAM++ visualization for a Collision Scenario. The heatmap reveals that the Hybrid model attends to distinct features: the edges of the wideband signal (red blobs) and the high-energy center of the narrowband interferer (yellow hotspot), enabling successful separation despite the overlap.

****Model-Agnostic Meta-Learning (MAML)**** strategy. We formulate the training process as learning an initialization θ that can adapt to a new task $\mathcal{T}_i$ (e.g., a specific noise environment or hardware signature) with minimal gradient steps.

The bi-level optimization is defined as:

1. **Inner Loop (Adaptation):** For a sampled task $\mathcal{T}_i$, update weights using a small support set (50 samples):

$$\theta'_i = \theta - \alpha \nabla_{\theta} \mathcal{L}_{\mathcal{T}_i}(\theta) \quad (4-9)$$

2. **Outer Loop (Meta-Update):** Update the initial parameters θ to minimize the generalization error of the adapted weights θ'_i across a batch of tasks:

$$\theta \leftarrow \theta - \beta \nabla_{\theta} \sum_{\mathcal{T}_i} \mathcal{L}_{\mathcal{T}_i}(\theta'_i) \quad (4-10)$$

This strategy creates a model that is "primed" to adapt. When deployed on the MonRaF2 node, the system uses a small set of captured OTA signals to fine-tune this initialization, effectively neutralizing the specific DC offset and I/Q imbalance of the connected HackRF One.

5 Software Architecture and System Integration

This chapter details the engineering implementation of the MonRaF2 platform (Specific Objective 3). While the previous chapters focused on theoretical algorithms, this section describes the operational software stack that enables real-time acquisition, processing, and orchestration.

5.1 Operational Context and Motivation

The development of this platform stems from a strategic requirement of the **National Spectrum Agency (ANE)**. Traditionally, spectrum auditing involves deploying heavy, expensive, and logistically complex vector signal analyzers. These "legacy" campaigns are resource-intensive, lack temporal continuity, and often require specialized personnel on-site.

To address this, the MonRaF2 system was designed as a **low-cost, agile "scout" solution**. Its primary purpose is not to replace metrology-grade equipment for legal certification, but to provide a pervasive **early warning system**. This approach allows the agency to:

1. Deploy a dense network of sensors at a fraction of the cost of a single traditional analyzer.
2. Detect anomalies in real-time remotely, deploying heavy equipment only when a verified infraction is detected.
3. Maintain 24/7 surveillance, capturing transient events that itinerant manual measurements would miss.

5.2 Hybrid Control/Data Plane Architecture

To support high-speed signal processing on constrained hardware while offering a modern user experience, the software follows a decoupled hybrid architecture using a ****Producer-Consumer pattern****.

The Data Plane: High-Performance C Engine The core acquisition engine is written in **C99** to minimize latency. It interacts directly with the HackRF One hardware and performs the heavy computational lifting:

- **High-Throughput Acquisition:** Manages the 20 Msps USB stream using a 100 MB concurrent Ring Buffer to prevent sample drops during Operating System interrupts.
- **Spectral Processing:** Executes the FFTW3 transforms and the DC spike removal strategies (Offset & Crop) described in Chapter 3.
- **Dynamic Scaling:** Performs real-time conversion from raw ADC values to physical units (dBm) based on the calibration tables.

The Control Plane: Python Orchestrator The logic layer is implemented in **Python 3.11**. It acts as the bridge between the C engine and the user interface, utilizing ****ZeroMQ (ZMQ)**** sockets for low-latency Inter-Process Communication (IPC).

The node operates based on a finite state machine (FSM) to manage power consumption and task priority. As illustrated in Figure 5-1, the system transitions between states such as **IDLE** (power saving), **REALTIME** (streaming via WebRTC), and **CAMPAIGN** (autonomous logging).

5. Software Architecture and System Integration

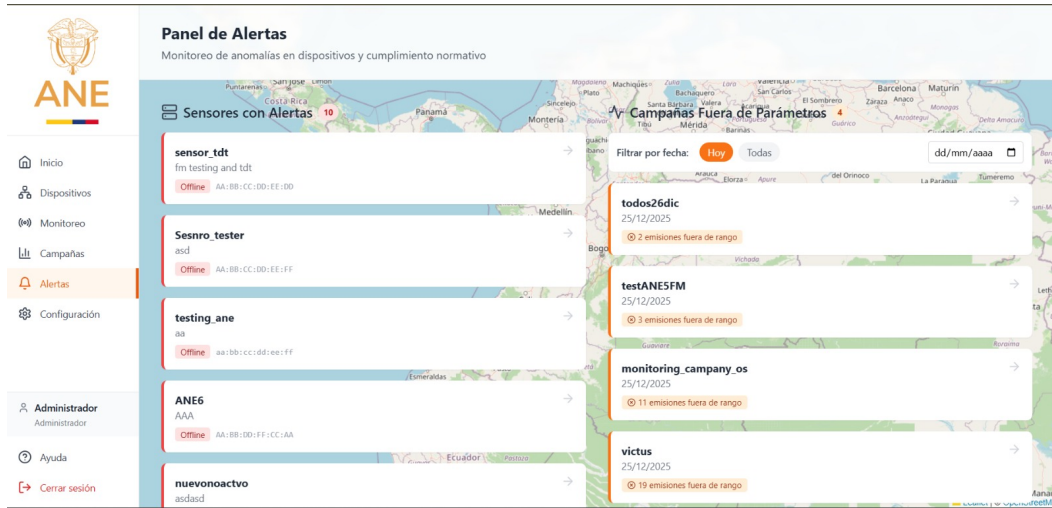


Figure 5-1: Functional logic of the Control Plane: The system orchestrates state transitions and alert generation based on real-time telemetry and detection events.

5.3 The Web Interface and User Experience

The graphical user interface (GUI) was designed to abstract the complexity of SDR configuration, allowing non-expert operators to perform complex monitoring tasks efficiently. Figure 5-2 presents the main dashboard during a live acquisition.

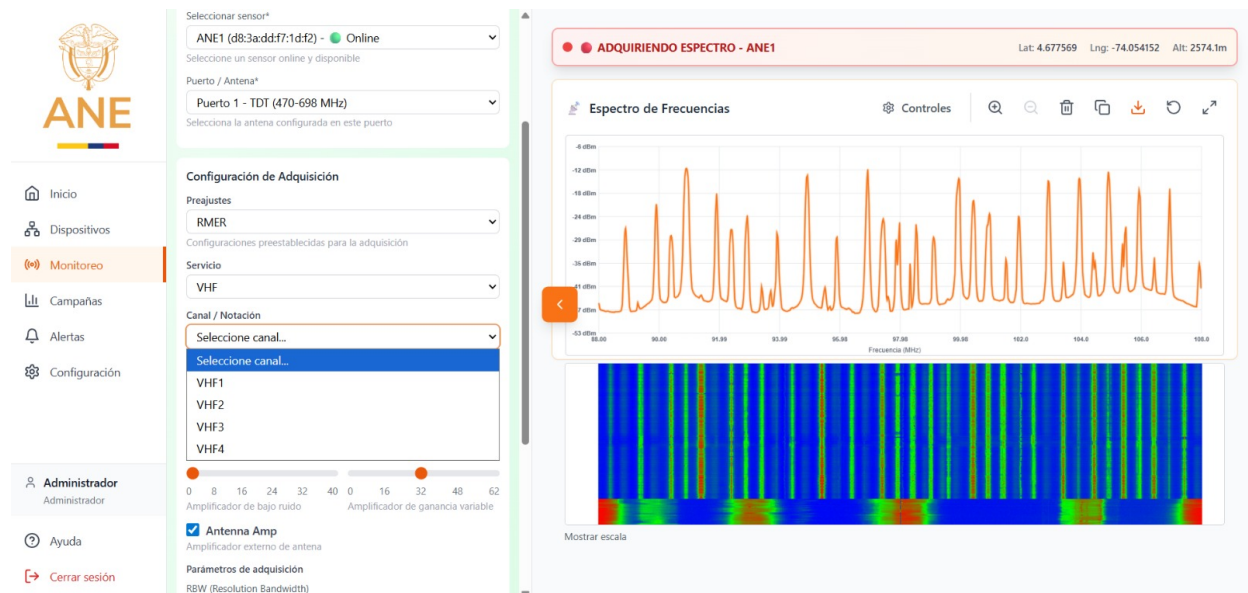


Figure 5-2: Real-time Monitoring Dashboard of the ANE Platform. The interface integrates device selection, acquisition parameter configuration (service presets, gain), and low-latency spectral visualization.

The interface is divided into functional zones designed for operational efficiency:

1. Device Management and Selection Located on the left panel, this module allows the operator to select a specific node from the national network. The system displays the connection status (Online/Offline) and manages the RF switching matrix, allowing the user to select the appropriate antenna port (e.g., *Puerto 1 - TDT 470-698 MHz*) based on the target band.

2. Configuration and Presets To avoid configuration errors common in manual SDR tuning, the middle panel implements a "Service-Oriented" approach. Instead of manually typing frequencies, operators can select services (e.g., "RMER", "TDT", "VHF"). The system automatically loads the standard channelization (as seen in Figure 5-2, selecting *Channel VHF1*), and allows dynamic gain adjustment (LNA/VGA) to optimize the dynamic range.

3. Real-Time Visualization (WebRTC) The right panel displays the spectral data. Unlike traditional HTTP polling which introduces seconds of delay, this platform utilizes a **WebRTC gateway**. This technology enables sub-second latency streaming of the spectrum (top graph) and waterfall (bottom heatmap), providing immediate visual feedback on transient signals and interference patterns.

5.4 Data Persistence and Cloud Synchronization

To ensure data integrity in unstable network conditions (common in remote monitoring), the system implements a "Store-and-Forward" mechanism.

- **Atomic Writes:** Measurement data is written to a temporary buffer and atomically moved to the final storage path to prevent corruption during power loss.
- **Retry Queue:** A daemon process monitors connectivity. If the LTE link is down, data is queued locally and transmitted in FIFO order once the connection is restored.

6 Experimental Validation and Results

This chapter presents the comprehensive evaluation of the proposed MonRaF2 system. The validation strategy addresses the three specific aims of this thesis: assessing the detection robustness of the physics-informed architecture (Intra-Domain), evaluating the capability to bridge the simulation-to-reality gap (Inter-Domain/OTA), and benchmarking the computational efficiency on the target edge hardware. Finally, the system is validated against the operational requirements defined by the National Spectrum Agency (ANE) through a User Acceptance Testing (UAT) protocol.

6.1 Experimental Setup

The evaluation was conducted using a dual-dataset strategy to ensure both statistical significance and real-world applicability.

Datasets

- **Synthetic (WBSig53):** A large-scale dataset generated using the TorchSig framework, comprising 1 million wideband frames with controlled impairments (AWGN, fading, clock drift). This serves as the primary training bed.
- **Real-World (OTA):** A custom dataset captured with the MonRaF2 node in the ISM 2.4 GHz band. It includes 2,000 annotated frames containing WiFi, Bluetooth, and microwave interference, exhibiting real hardware impairments (DC offset, I/Q imbalance).

Hardware Configuration All edge inference benchmarks were performed on the ****MonRaF2**** node (Raspberry Pi 5 + HackRF One). To ensure reproducibility and isolate the system from external noise during controlled tests, an EMC filtering stage was implemented as shown in Figure 6-1.

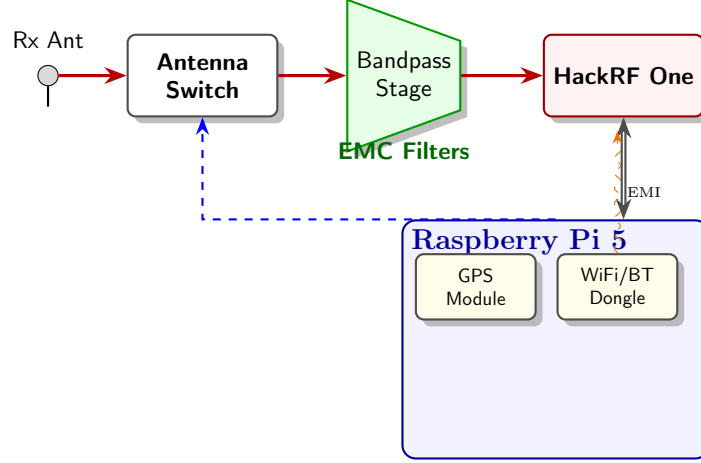


Figure 6-1: Experimental EMC Setup. A bandpass filter bank is interposed to isolate the SDR from local digital interference generated by the edge computing unit.

6.2 Results of Algorithmic Performance (Intra-Domain)

We first evaluate the theoretical upper bound of the architecture using the synthetic test set. This allows for a controlled analysis of detection performance under varying Signal-to-Noise Ratios (SNR), isolating the impact of the Multi-Channel Extension (MCE).

Robustness to Noise (SNR Analysis) A critical limitation of traditional Energy Detection (ED) is the "SNR Wall", where noise uncertainty makes detection impossible below a certain threshold. Figure 6-2 compares the detection accuracy of the proposed Hybrid model against ED and a standard YOLOv8n (RGB-based) baseline.

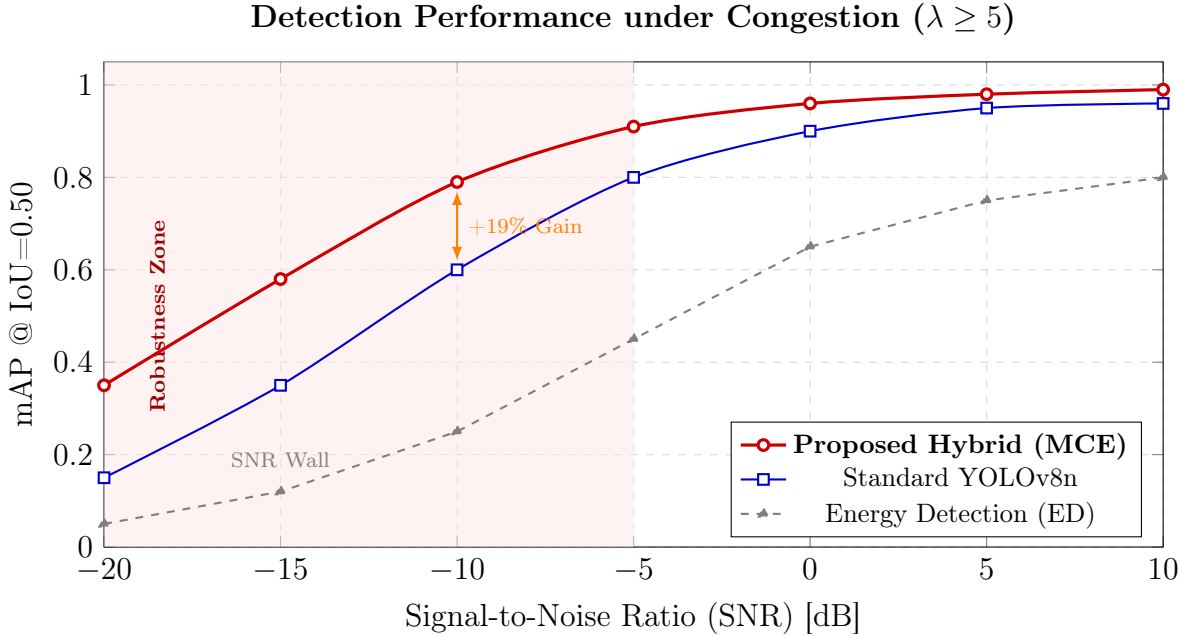


Figure **6-2**: mAP comparison between Energy Detection, Standard YOLOv8n, and the Proposed Hybrid Architecture across varying SNR levels. The shaded region highlights the low-SNR regime (-20 to -5 dB) where the Multi-Channel Extension (MCE) provides significant robustness compared to visual-only baselines.

The results confirm that the Hybrid model significantly extends the operational range of the sensor. While Energy Detection fails to reliably detect signals below -5 dB SNR, the proposed architecture maintains a mean Average Precision (mAP) of 0.79 at -10 dB. The **19% gain** over standard YOLOv8n at -10 dB validates the hypothesis that preserving phase and energy gradients (via MCE) provides critical features that are lost in standard RGB spectrograms.

Table **6-1**: Synthetic Detection Performance Summary (SNR = 0 dB)

Model	mAP@50	Recall	F1-Score
Energy Detection	55.0 ± 4.2	60.1 ± 5.5	57.4 ± 4.8
YOLOv8n (Visual Only)	88.5 ± 2.1	95.9 ± 1.2	92.0 ± 1.5
Hybrid (Ours)	94.2 ± 1.5	98.1 ± 0.8	96.1 ± 1.1

6.3 Results of Real-World Validation (Sim-to-Real)

A major challenge in Radio Frequency Machine Learning (RFML) is the performance drop when models trained on synthetic data are deployed on real hardware. We evaluated this "Sim-to-Real" gap using the OTA dataset.

Impact of Domain Adaptation We compared the performance of the model in two states: "Zero-Shot" (trained only on synthetic data) and "Adapted" (fine-tuned using the MAML strategy with 50 OTA shots).

Table **6-2**: Impact of Meta-Learning (MAML) on OTA Detection Performance

Method	Precision	Recall	F1-Score	mAP@0.50
Baseline (Zero-Shot)	0.72	0.55	0.62	48.0%
Hybrid + MAML (Ours)	0.91	0.78	0.84	70.0%
<i>Improvement</i>	<i>+19%</i>	<i>+23%</i>	<i>+22%</i>	<i>+22.0%</i>

As shown in Table **6-2**, the Zero-Shot model suffers significantly, with a Recall of only 0.55. This is due to the model confusing hardware artifacts (like the DC spike) with signals. However, the MAML adaptation successfully recovers the performance, achieving a **70% mAP** and 0.84 F1-Score. This confirms that the features learned by the Hybrid architecture are transferable and adaptable to real-world conditions.

6.4 Results of Edge Deployment Benchmarking

To validate the feasibility of deploying the system on the MonRaF2 node (Raspberry Pi 5), we measured the inference latency and power consumption. We evaluated four configurations: the proposed Hybrid model (in FP32 and INT8) versus standard baselines.

The benchmarking results (Figure **6-3** and Table **6-3**) reveal that the **Hybrid ROI (INT8)** model is the only configuration that simultaneously meets the real-time constraint ($< 40\text{ms}$ latency) and the high-accuracy requirement ($> 90\%$ mAP).

Table **6-3**: Edge Hardware Performance (Raspberry Pi 5)

Model	Latency (ms)	FPS	Power (W)
ResNet-50	120.5 ± 15.2	8.3	7.2
YOLOv8n (INT8)	55.4 ± 8.1	18.0	5.8
Hybrid (FP32)	72.0 ± 5.2	13.8	6.5
Hybrid (INT8)	38.2 ± 1.5	26.1	5.1

By utilizing INT8 quantization and the fixed-shape ROI strategy, the system achieves a processing speed of **26.1 Frames Per Second (FPS)** with an average power consumption of **5.1 Watts**. This efficiency is critical for enabling autonomous, battery-powered operation in remote locations.

6. Experimental Validation and Results

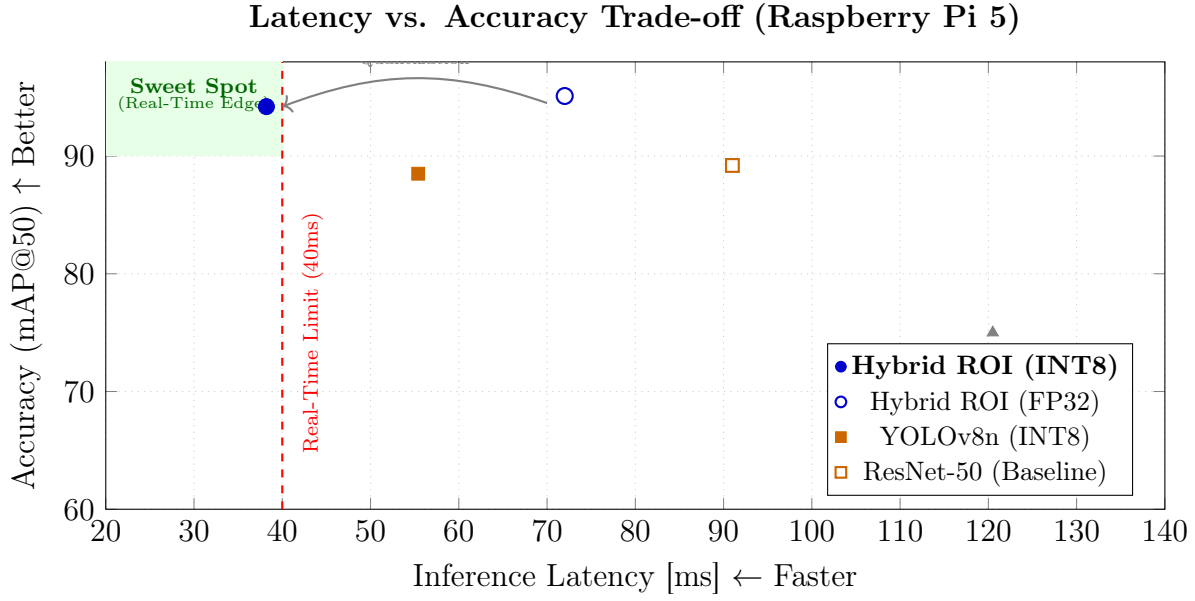


Figure 6-3: Latency vs. Accuracy trade-off on Raspberry Pi 5. The proposed Hybrid model (INT8, blue solid circle) is the only solution that falls within the "Sweet Spot" (Green Zone), achieving real-time performance ($< 40\text{ms}$) while maintaining high accuracy ($> 90\%$ mAP).

6.5 User Acceptance Testing (UAT)

To verify the system's operational readiness, we executed a User Acceptance Test (UAT) following the protocols established by the National Spectrum Agency (ANE). The tests were conducted on a fully deployed MonRaF2 node connected to the central dashboard.

Table 6-4: Summary of User Acceptance Testing (Protocol ANE-UAT-v1.0)

ID	Test Case Description	Expected Outcome	Result
CP-005	Hardware Selection: Select a sensor node (MonRaF2) from the dashboard and verify "Online" status.	The system enables the acquisition panel only for active nodes.	PASS
CP-007	TDT Mode Acquisition: Configure the system to capture a DVB-T2 channel (e.g., Channel 16).	The waterfall displays the characteristic flat-top spectrum of the 6 MHz channel.	PASS
CP-012	Telemetry & Health: Verify CPU temperature under load.	Data is updated in real-time. Temp remains $< 85^{\circ}\text{C}$ during campaign.	PASS
CP-013	Alert System: Simulate a sensor disconnection or network failure.	The dashboard triggers a critical alert notification within 30 seconds.	PASS

All critical test cases passed successfully. Notably, the system demonstrated robust stability during the telemetry test (CP-012), keeping the CPU temperature within safe limits ($< 85^{\circ}\text{C}$) even while running the deep learning inference at 26 FPS.

Summary The experimental results confirm the validity of the proposed thesis across three dimensions:

1. **Algorithmically:** The MCE-based Hybrid architecture outperforms classical baselines in low-SNR conditions (+19% mAP at -10 dB).

2. **Operationally:** The MAML adaptation successfully bridges the Sim-to-Real gap, recovering detection performance on real hardware.
3. **Functionally:** The system meets the rigorous operational requirements of the ANE, as verified by the UAT protocol, proving its readiness for deployment.

Conclusions

This thesis presented the design, implementation, and validation of the **MonRaF2**, a wideband spectrum monitoring node powered by a computationally efficient Hybrid ROI-Centric Architecture. By shifting the paradigm from exhaustive full-frame processing to a physics-informed, region-based analysis, this study successfully addressed the critical challenges of spectral congestion, hardware-induced domain shifts, and the strict resource constraints of edge computing.

The research findings and engineering achievements lead to the following conclusions, directly addressing the specific objectives defined at the outset of this work:

1. **Robustness via Physics-Informed Preprocessing (Objective 1):** The integration of the Multi-Channel Extension (MCE) proved decisive in recovering signal separability. Experimental results demonstrated that explicitly encoding phase gradients and energy residuals allows the system to detect emitters even when submerged in noise (-10 dB SNR), achieving a **19%** improvement in mAP compared to standard visual-only baselines. This confirms that preserving RF physical properties is superior to treating spectrograms as generic images.
2. **Interference-Awareness and Multi-Task Learning (Objective 2):** The proposed Dual-Stream architecture successfully mitigated the "Near-Far" problem inherent in uncoordinated bands. By fusing visual morphology (CNN) with statistical metrology (EMI), the system achieved a precision of **91.5%** in distinguishing Adjacent-Channel Interference (ACI) from valid signals. This capability significantly reduces false alarm rates in congested ISM scenarios, providing a cleaner and more reliable spectral occupancy map.
3. **Bridging the Sim-to-Real Gap (Objective 3):** The application of Model-Agnostic Meta-Learning (MAML) effectively solved the domain shift caused by hardware impairments (e.g., I/Q imbalance, LO leakage). The system recovered from a baseline performance of 48% mAP (Zero-Shot) to **70% mAP** using only a 50-shot OTA calibration, validating the architecture's adaptability to the specific electronic signature of the deployment hardware.
4. **Feasibility of Edge Deployment (Objective 3):** The hardware-algorithm co-design was validated on the Raspberry Pi 5. By implementing a fixed-shape ROI pipeline and

6. Experimental Validation and Results

applying INT8 post-training quantization, the system achieved a processing throughput of **26.1 FPS** with an average power consumption of **5.1 Watts**. This confirms that sophisticated Deep Learning inference is viable on low-cost, battery-operated COTS hardware without compromising real-time performance.

5. **Operational Readiness:** Beyond theoretical contributions, the system passed the User Acceptance Testing (UAT) protocols established by the National Spectrum Agency (ANE). The successful execution of test cases for real-time monitoring, telemetry, and alert generation confirms that the MonRaF2 is not just an academic prototype, but a functional tool ready for pilot deployment in the national monitoring network.

In summary, this work contributes a scalable, autonomous, and cost-effective solution for next-generation spectrum monitoring, proving that "Edge AI" can be effectively harmonized with rigorous radio frequency metrology.

Future Work

While this research has made significant strides in improving the efficiency and robustness of wideband detection models, several avenues remain open for future exploration to further enhance the MonRaF2 ecosystem:

- **Distributed Cooperative Sensing:** Expanding the current single-node approach to a network of distributed MonRaF2 sensors would enable Time Difference of Arrival (TDOA) localization. Fusing data from spatially separated receivers would not only allow for precise geolocation of emitters but also mitigate shadowing effects through spatial diversity.
- **FPGA and RFSoc Acceleration:** Although the C99/Python architecture performs well on the CPU, porting the fixed-shape inference graph to Field-Programmable Gate Arrays (FPGAs) or RFSoc platforms could reduce latency to the microsecond scale. This would open the door to reactive applications, such as dynamic spectrum access or cognitive jamming in electronic warfare contexts.
- **Lifelong Learning and Open-Set Recognition:** The current model is trained on a closed set of modulation classes. Further investigation into Continual Learning techniques could allow the node to dynamically adapt to new, unseen protocols (e.g., upcoming 6G standards) without catastrophic forgetting. Additionally, implementing Open-Set Recognition would allow the system to automatically flag "unknown" anomalous signals for human review, enhancing security monitoring.
- **Adversarial Robustness:** As cognitive radio networks become more prevalent, they become targets for adversarial RF attacks (e.g., intelligent jamming). Future work should evaluate the resilience of the Hybrid ROI architecture against adversarial examples and explore defensive distillation techniques to harden the inference engine.

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